AKADEMIKA

DCL-BBT

BASE BAND DIGITAL TRANSMISSION SYSTEM KIT

EXPERIMENTAL MANUAL

….A MESSAGE FROM

Today's system designers are faced with tomorrow's problems. ADVANCED DIGITAL COMMUNICATION is one of the important subject need to teach while learning electronics.
It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

OUR MOTTO;

-Light years ahead – refers to leadership.

As leaders in our industry in India, we are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that we have complete, truly innovative & superior training products. We are so committed to quality that we back our products with a complete comprehensive warranty.

AKADEMIKA LAB SOLUTIONS

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SAFETY RULES

Carefully follow the instructions contained in this manual as they provide you with important points on safety during the installation, use and maintenance. Keep this manual always with you for easy reference.

Arrange all accessories in order after unpacking, so that its integrity is checked with respect to its packing list. Also ensure that no visible damage as such appear on any accessories.

Before connecting the power supply to the kit, be sure that the jumpers and connecting chords are connected correctly as per experiment.

In DCL-BASE BAND kit, signals are to be monitored with an oscilloscope as explained in the manual. For observation of signals on oscilloscope either use X10 (Attenuation Probe) or 180Ω resistance in series with normal oscilloscope probe.

These kits must be employed only for the use for which it has been conceived, i.e. as educational equipment, and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of any fault or malfunctioning in the trainer kit, turn off the power supply and do not tamper the kit. In case servicing is required, contact the service center for technical assistance.

The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity**.**

WARRANTY

These kits are warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like connecting chords, Crystals, etc. and other imported items.

This warranty is subject to the following conditions and limitations. The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product is, on Akademika Lab Solutions sole judgment. The defective product will be replaced with new one or repaired without charge or with charge.

In the warranty period if the service is needed, purchaser should get in touch with the service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of kits has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on Akademika Lab Solutions judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster Akademika Lab Solutions. is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up-gradation of products, be sure to contact our service center or sales outlet.

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TECHNICAL SPECIFICATIONS

Noise Generator
White noise source output
Amplitude : 0 to 4V.

PRBS Generator
16 bit switch selectable.
Clock rate : (16, 32, 64,128,256,512) KHz, 1.024MHz.

Bit Error Rate Meter : 4 Digit counter with 7 segment display
Display : 4 Digit 7 segments. Count up to 9999.

Digital Modulation
Modulation technique : Pulse amplitude modulation.
Duty cycle : 50%.

Transmitter Filter : 1st order Butterworth filter Selectable switch With five different bands

Receiver Filter : 1st order Butterworth filter Selectable Switch with five different bands

Coding Operation
Scrambler : 16 bit data pattern.
Unscramble : 16 bit data pattern.

Inter Connection : 2mm banana socket.

Test Points : 30 nos. test points are provided on board to Observe intermediate signals

Switch Faults : 8 switch faults are provided on board to study Different effects on circuit

Power Supply : -12V, +12V, +5V, GND.

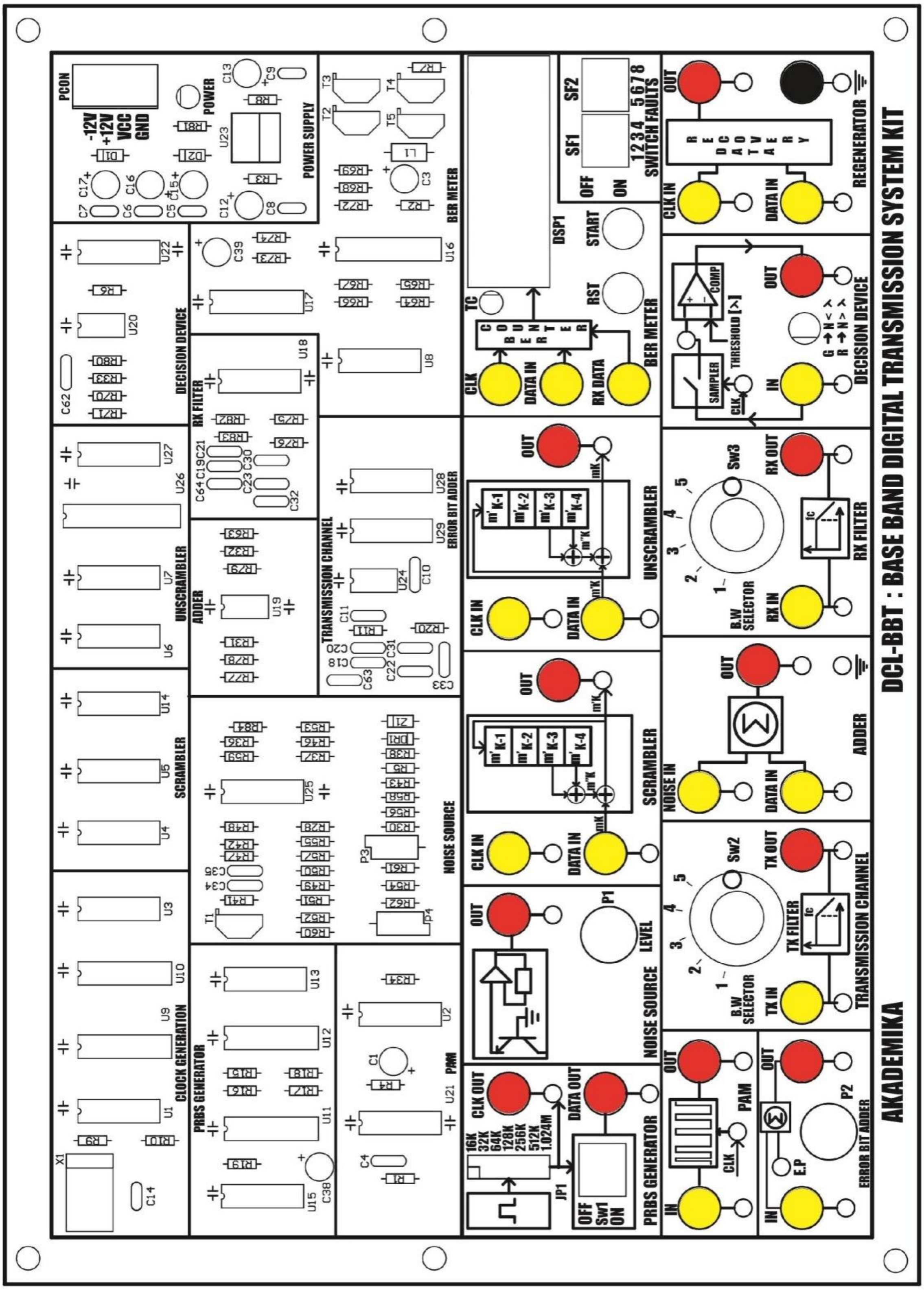
Accessories

	DCL-BBT		Quantity
1	Short Links PCS2 Patch cord	Male To Male	10
2	ADCL-04 EXPT Manual		1
3	Power Supply		1
4	Jumper Caps Small		5
5	Power cord		1

FEATURES

- On board clock generator:
- PRBS generator.
- Noise source.
- Bit error rate measurement.
- Eye pattern.
- Scrambler.
- Unscramble.
- PAM.
- Band limiting.
- Provision for interfacing with other DCL-Base Band lab kits.
- Switch Faults.

FUNCTIONAL BLOCK



INTRODUCTION

AKADEMIKA's DCL- Base Band Digital Transmission System kit is an Advanced Digital Communication Trainer kit. A single board unit designed to learn the fundamentals of base band digital transmission system including properties of transmitter and receiver filters, Pulse Amplitude Modulation of binary data, Band Limiting, observation of Eye Pattern, Bit Error Rate measurement, study of coding operation like Scrambling and Unscrambling.

Band Digital Transmission

Baseband transmission in order to emphasize the generic concepts and problems of digital communication, with or without carrier modulation, and to bring out the differences between digital and analog transmission Following an overview of digital signals and systems, we'll analyze the limitations imposed by additive noise and transmission bandwidth. The task of digital communication system is to transfer a digital message from the source to the destination. But finite transmission bandwidth sets an upper limit to the symbol rate, and noise causes error to appear in the output message. Thus signaling rate and error probability play roles in digital communication similar to those of bandwidth of a signal to noise ratio in analog communication.

Digital data have a broad spectrum with a significant low frequency content. Baseband transmission of digital data therefore requires the use of a low-pass channel with a bandwidth large enough to accommodate the essential frequency content of the data stream. Typically, however, the channel is dispersive in that its frequency response deviates from that of an ideal low-pass filter. The result of data transmission over such a channel is that each received pulse is effected somewhat by adjacent pulses, thereby giving rise to a common form of interference called inter-symbol interference (ISI). Inter-symbol interference is the major source of bit errors in the reconstructed data stream at the receiver output correct for it control has to be exercised over the pulse shape in the overall system.

Another source of bit errors in a baseband data transmission system is the ubiquitous channel noise. Naturally noise and ISI arise in the system simultaneously.

Digital Signals and Systems

A digital message is nothing more than an ordered sequence of symbols produced by a discrete information source. The source draws from an alphabet of $M \geq 2$ different symbols, and produces output symbols at some average rate *r.* For instance, a typical computer terminal has an alphabet of $M \approx 90$ symbols, equal to the number of character keys multiplied by two to account for the shift key. When you operate the terminal as fast as you can, you become a discrete information source producing a digital message at a rate of perhaps $r \approx 5$ symbols per second. The computer itself works with just $M = 2$ internal symbols, represented by a LOW and HIGH electrical states. We usually associate these two symbols with the *binary digits* 0 and 1, known as bits for short.

PRBS Sequence

In ADCL-04, PRBS sequence is generated by using a 4-bit right shift register whose feedback is completed by the EX-OR gate.

Let initially 1001 be the 4 bit switch setting on the SW1.

Clock States	D1	D2	D3	D4	
	A	B	C		
1	1	0	0	1	1
2	1	1	0	0	0
3	0	1	1	0	1
4	1	0	1	1	0
5	0	1	0	1	1
6	1	0	1	0	1
7	1	1	0	1	1
8	1	1	1	0	1
9	1	1	1	1	0
10	0	1	1	1	0
11	0	0	1	1	0
12	0	0	0	1	1
13	1	0	0	0	0
14	0	1	0	0	0
15	0	0	1	0	1
16	1	0	0	1	1

Thus the sequence repeats constantly with a period corresponding to 16 clock states.

Length of sequence = 2^4 =16

Now the Pseudo Random Sequence pattern is C = 1010111100010011

Digital PAM Signals

Digital message representation at baseband commonly takes the form of an amplitude modulated pulse train. We express such signals by writing

$$X(t) = \sum_{K} a_k p(t-kD) \qquad (1)$$

where the modulating amplitude a_k represents the *k*th symbol in the message sequence, so the amplitudes belong to a set of M discrete values. The index k ranges from $-\infty$ to $+\infty$ unless otherwise stated. Equation (1) defines a digital PAM signal as distinguished from those rare cases when pulse-duration or pulse-position modulation is used for digital transmission.

The un-modulated pulse *p(t)* may be rectangular or some other shape, subject to the conditions. The condition ensures that we can recover the message by sampling *x(t)* periodically.

Transmission Limitations

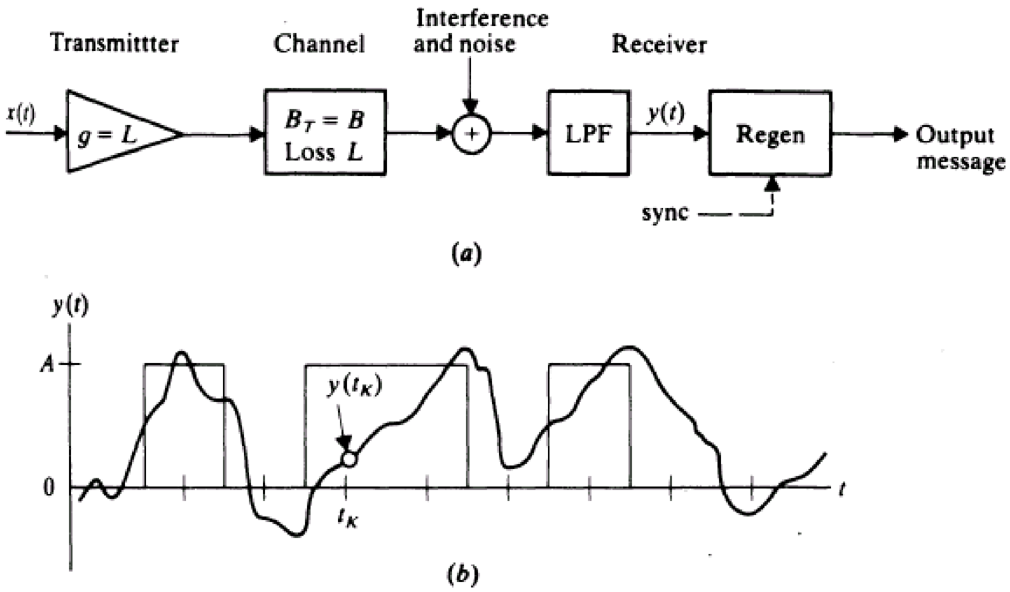


Figure 1 (a) baseband transmission system; (b) signal-plus-noise waveform

Now consider the linear baseband transmission system diagrammed in Fig 1.a.We'll assume for the convenience that the transmitting amplifier compensates for the transmission loss, and we'll lump any interference together with the additive noise. After low pass filtering to remove out-of-band contaminations, we have the signal-plus-noise waveform

$$y(t) = \sum_k a_k p(t - t_d - kD) + n(t)$$

Where,

t_d is the transmission delay.

p(t) is the pulse shape with transmission distortion.

The Fig 1.b.illustrates that *y(t)* might look like when *x*(t) is the uni-polar NRZ signal. Recovering the digital message from *y(t)* is the task of the *regenerator*. An auxiliary synchronization signal may help the regeneration process by identifying the optimum sampling times

$$t_k = KD + t_d$$

If p(0) = 1 then,

$$y(t_k) = a_k + \sum_{k \neq K} a_k p(KD - kd) + n(t_k) \qquad \text{(a)}$$

Whose first term is the desired message information The last term of the equation (a) is the noise contamination at t_k, while the middle term represents cross talk or spill over from other signal pulses a phenomenon given the descriptive name inter-symbol interference (ISI). The combined effects of noise and ISI may result in *errors* in the regenerated message. For instance, at the sampled time t_k indicated in the Fig 1.b., $y(t_k)$ is closer to zero even though $a_k = A$.

We know that if *n(t)* comes from a white noise source, then its mean square value can be reduced by reducing the bandwidth of the LPF. We also know that low pass filtering causes pulses to spread out, which would increase the ISI. Consequently, a fundamental limitation of digital transmission is the relationship between ISI, bandwidth, and signaling rate.

Band Limiting

The displays you have seen on the oscilloscope are probably as you would have expected them to be! That is, either 'HI' or 'LOW' with sharp, almost invisible, transitions between them. This implies that there was no band limiting between the signal and the viewing instrument. If transmitted via a low pass filter, which could represent a band limited (baseband) channel. We must use band limited pulse specially shaped to avoid ISI.

CIRCUIT DESCRIPTION

- **Clock Generator**

Refer to the circuit diagram as shown later in this manual sheet 1 of 8 of DCL-BBT.This block consists of Crystal oscillator, which generates a 1.024MHz clock, from IC 74HCT14 and is supplied to U9 (IC 74HC193) from which rest of the required clock signals are derived.

- **PRBS Generator**

Refer to the circuit diagram as shown later in this manual sheet 1 of 8 of DCL-BBT.It is a 14-bit Pseudo Random sequence used as a data to be transmitted along the baseband transmission system. A 4-bit DIP switch SW1 is used to change the pattern of PRBS sequence. A 4-bit shift register U12 (74LS195) generates the pseudo Random sequence, which is a unique pattern that has minimum correlation with the data. Different clock frequencies starting from minimum 16 KHz to 1.024 MHz is fed to the PRBS generator to generate the PRBS pattern of different frequencies. The PRBS generator is a 4-bit shift register with an EX-OR U13 (IC74HC86) gate at the feedback.

- **Noise Source**

Refer to the circuit diagram as shown in this manual sheet 2 of 8 of DCL-BBT. The noise source consists of a transistor T1 (BC547) whose base is grounded, collector is left open and the emitter is given supply voltage. In this biasing condition it produces a noise in the order of some tens of mV. The noise is amplified by two amplifier stages U25 (TL084) with a potentiometer P1 for level regulation.

- **Bit Error Rate Meter**

Refer to the circuit diagram as shown later in this manual sheet 5 of 8 of DCL-BBT. Transmitted data and received data bits are passed through buffer U15 (74HCT04) to D-flip-flop U8 (74HCT74). Output of the D-flip-flop is EX-ORed. The error pulse generated through PAL and is given to U16 (74C926), which is a 4-digit counter with multiplexed 7 segment output drivers. The error measured is displayed on the 4 digit 7 segment displays with a decimal value. The counter can be reset through Reset switch followed by start switch**.**

- **Pulse Amplitude Modulation Section**

Refer to the circuit diagram as shown later in this manual sheet 6 of 8 of DCL-BBT. This section provides the pulse modulated signal output at the PAM OUT Post. The Pulse Amplitude Modulation circuitry is built around the analog switch U21 (CD4016). The amplitudes of the regularly spaced rectangular pulses vary in direct proportion to the instantaneous sample values of continuous digital pulses.

- **TX Filter**

Refer to the circuit diagram as shown later in this manual sheet 6 of 8 of DCL-BBT.The transmitter filter consists of a first order low pass filter built around U18 (ICTL084) .The bandwidth of the filter is varied by selecting capacitors of different values (C18, C20, C22, C31, C33) using rotary switch SW3.

- **RX Filter**

Refer to the circuit diagram as shown later in this manual sheet 6 of 8 of DCL-BBT. The receiver filter consists of a first order low pass filter built around U18 (ICTL084) .The bandwidth of the filter is varied by selecting capacitors of different values (C19, C21, C23, C30, C32) using rotary switch SW4.

- **Error Bit Adder**

Refer to the circuit diagram as shown later in this manual sheet 6 of 8 of DCL-BBt. 555 timer is configured as astable multivibrator. An astable multivibrator, often called a *free-running* multivibrator, is a rectangular-wave-generating circuit. Unlike the monostable multivibrator, this circuit does not require an external trigger to change the state of the output, hence the name *free-running*. However, the time during which the output is either high or low is determined by the two resistors and a capacitor, which is externally connected to the 555 timer IC U29 adds these error pulses with data.

- **Adder**

Refer to the circuit diagram as shown later in this manual sheet 8 of 8 of DCL-BBT. Adder circuit is built around U19 (IC CA3140). It adds the noise and the data and gives a noisy data at the output of the adder, whenever the noise level is high.

- **Decision Device**

Refer to the circuit diagram as shown later in this manual sheet 8 of 8 of DCL-BBT. The decision device consists of a comparator circuit built around U20 (IC LM311). A threshold voltage is fixed using a voltage divider circuit. The detected output of the decision device is buffered using U2 (IC74HCT14) .The bi-color LED is used to detect the fine data and the noisy data with two different color indications namely green and red.

- **Recovered Data**

Refer to the circuit diagram as shown later in the manual sheet 8 of 8 of DCL-BBT. U3 (IC74HCT74) D-flip-flop is used to recover the original data.

- **Scrambler**

Refer to the circuit diagram as shown later in the manual sheet 4 of 8 of DCL-BBT. The data is shifted using U4 and U5 (IC74HCT74) a D- flip-flop. The D-flip-flop is clocked using an external clock. The shifted data and the original data are EX-ORed using U14 (IC74HC86). This “randomizes” the bit stream, which generates the scrambled data according to the given input data.

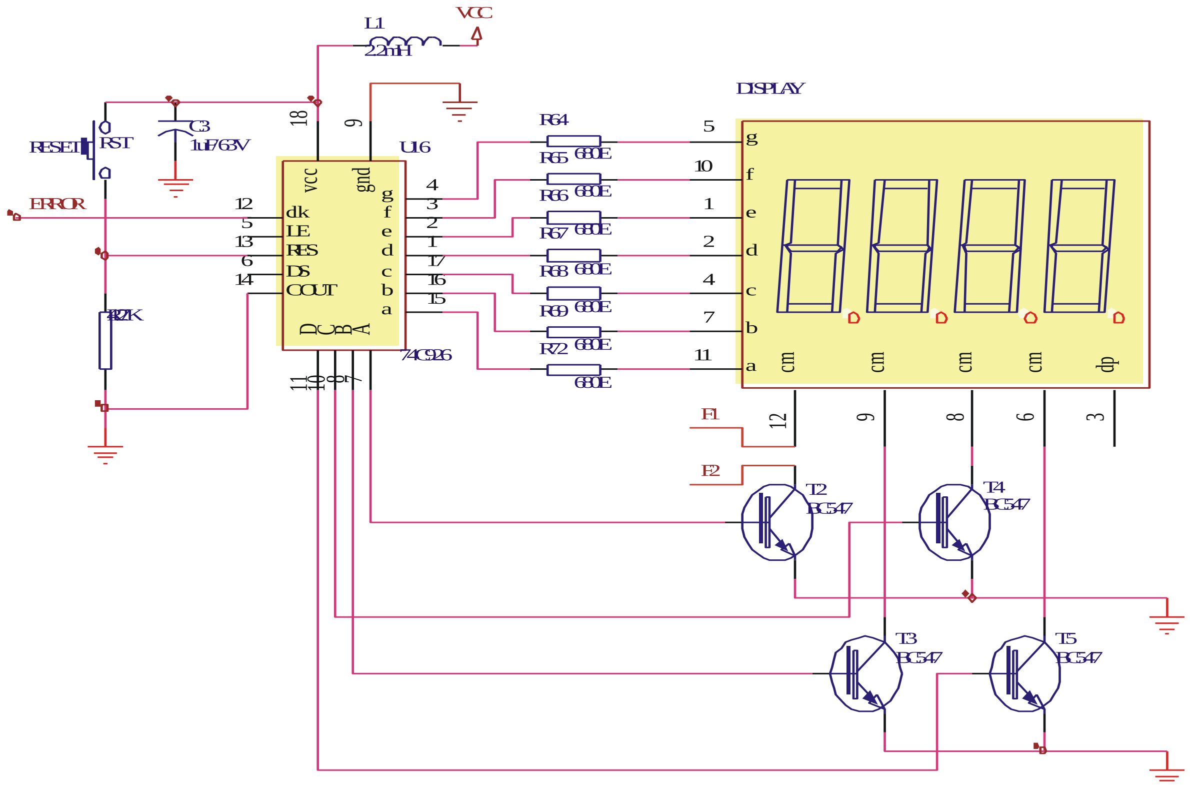
- **Unscrambler**

Refer to the circuit diagram as shown later in the manual sheet 4 of 9 of DCL-BBT. The scrambled data is shifted using U6 and U7 (IC74HCT74) a D- flip-flop. The D-flip-flop is clocked using an external clock. The shifted data and the original data are EX-ORed. Hence the scrambled data is unscrambled and the original data is recovered as per the input data.

SWITCH FAULTS

Switch fault section on DCL-BASE BAND provides 8 switches. These switches can be used to simulate fault conditions in various parts of the circuit. The faults are normally used one at a time. To ensure that the DCL-BASE BAND is fully operational, all switches should be set to OFF position before use.

1) Effect of Switch Fault 1(DCL-BBT) in the BIT ERROR RATE section



DESCRIPTION

This circuit measures the BER for a given transmission. The BER, or quality of the digital link, is calculated from the number of bits received in error divided by the number of bits transmitted. This is easily measured by means of a comparator in which the transmitted bits are matched in an XOR gate with the received bits. If the bits are alike at the XOR gate inputs when clocked in from the D flip-flop, the output is low.
If they are different, the XOR output goes high, causing an event count. A random character generator and digital noise source is used for these measurements.

PROCEDURE

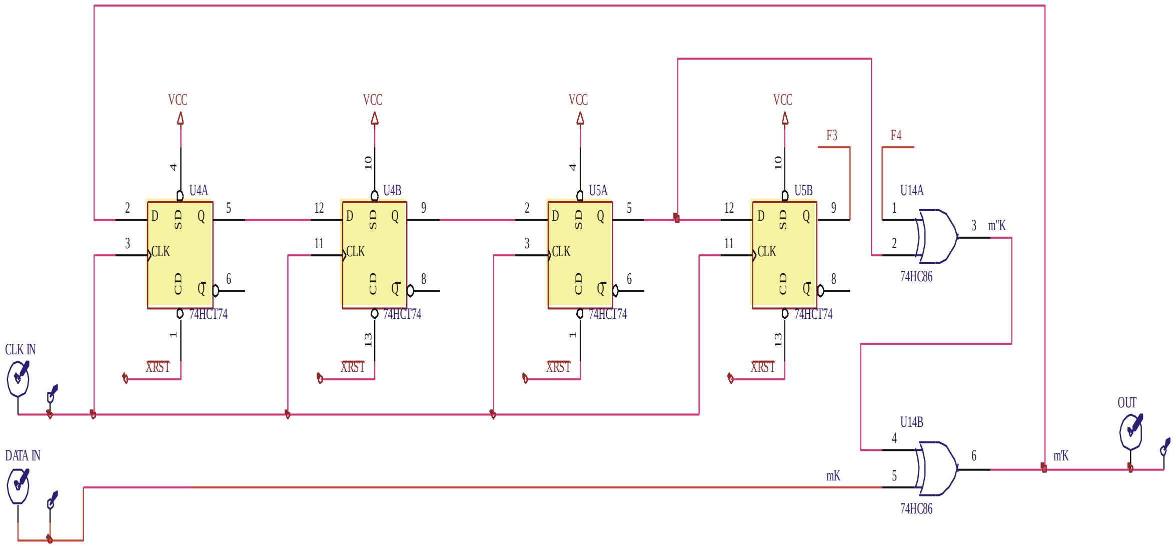
1. Connect power supply in proper polarity to the kit **DCL-BBT** and switch it on.
2. Keep the position of the switch **SW1** of the **PRBS GENERATOR** as shown in the Fig.5.1 for the generation of the PRBS pattern.
3. Select the **PRBS** clock frequency to **16 KHz** using jumper **JP1**.
4. Connect the PRBS **DATA OUT** post to the **IN** post of **ERROR BIT ADDER.**
5. Observe the noise pulses at **E.P.** by varying the pot **P2.**

6. Connect the **OUT** post of the Adder to the **RX DATA** post of the BER meter.
7. Keep the pot **P2** of the **ERROR BIT ADDER** at minimum position.
8. Connect the **CLK OUT** post to the **CLK** post of the BER meter.
9. Connect the **DATA OUT** post of **PRBS GENERATOR** to **DATA IN** post of the **BER METER**.
10. Press the **RST** switch to reset the counter to zero.
11. Press the **START** switch to start the counting.
12. Observe the error count on the display after 10 seconds upon pressing **START** switch.
13. Observe that when the pot is kept at minimum position the bit error count is very low. The error count will increase linearly as we increase the pot to the maximum (clockwise) position.

Action and Effect

The number of bit errors is dependent upon the amount of noise entering the system. When the switch **1** of **SF1** is put to ON position, the transistor T2 that drives the MSB of the 4-digit 7-segment display gets opened due to which the MSB does not glow.

2) Effect of Switch Fault 2(DCL-BBT) in the scrambler section



DESCRIPTION

This circuit is used for scrambling the data. The data is shifted using U4 and U5 (IC74HCT74) a D- flip-flop. The D-flip-flop is clocked using an external clock. The shifted data and the original data are EX-ORed using U14 (IC74HC86). This “randomizes” the bit stream, which generates the scrambled data according to the given input data.

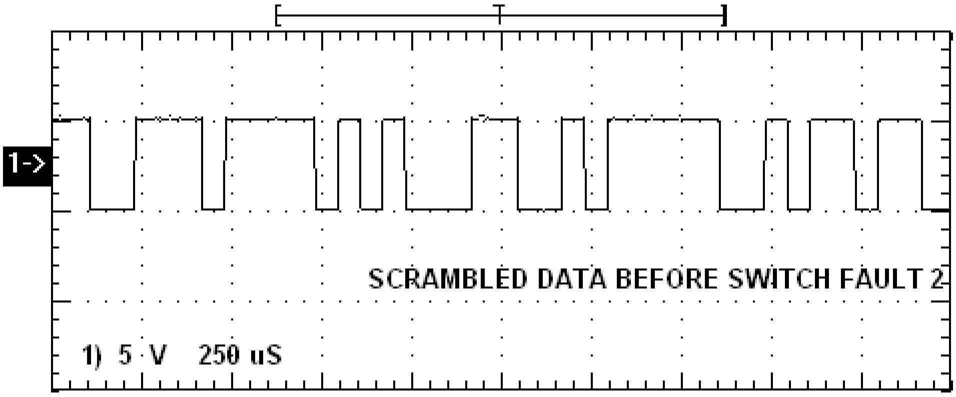
PROCEDURE

First keep switch **2** of **SF1** to OFF position.

1. Connect power supply in proper polarity to the kit **DCL-BBT** and switch it ON.
2. Keep the switch **SW1** as per the position shown in the block diagram fig.6.1 to generate the PRBS data pattern.

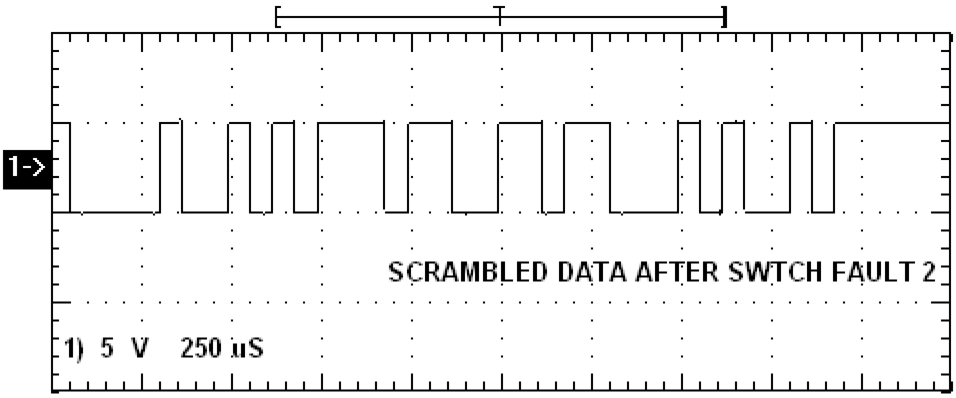
3. Connect the **CLK OUT** post of the PRBS generator to the **CLK IN** post of the Scrambler.
4. Connect the **DATA OUT** post to the **DATA IN** post of the Scrambler.
5. Observe the output data at the **OUT** post of the scrambler.

Waveform before Switch Fault



Now put the switch **2** of **SF1** in the Switch Fault to ON position and observe its effect.

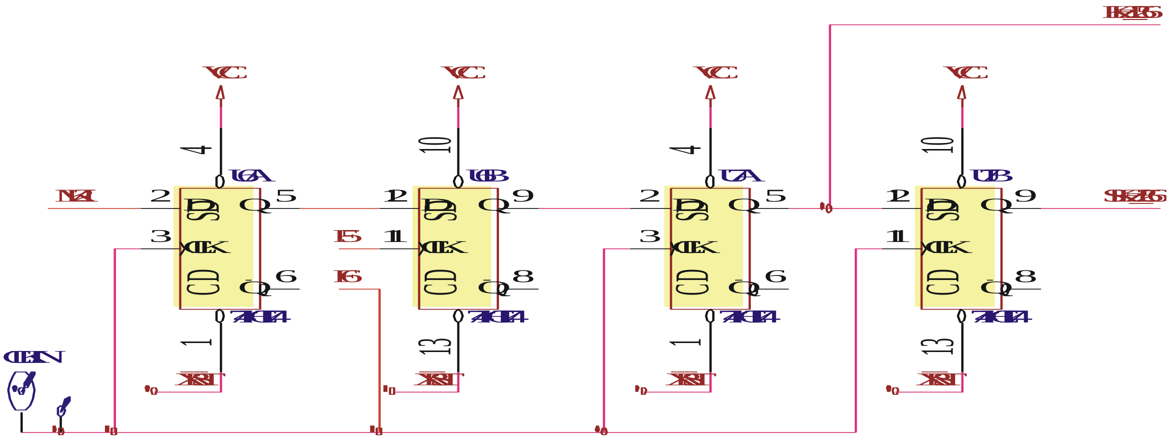
Waveform after Switch Fault



Action and Effect

When the switch **2** of **SF1** is put to ON position, the connection going to pin no.1 of U14 gets opened due to which there is loss in data. There is a loss in the output data of scrambler

3) Effect of Switch Fault 3 (DCL-BBT) in the unscramble section



DESCRIPTION

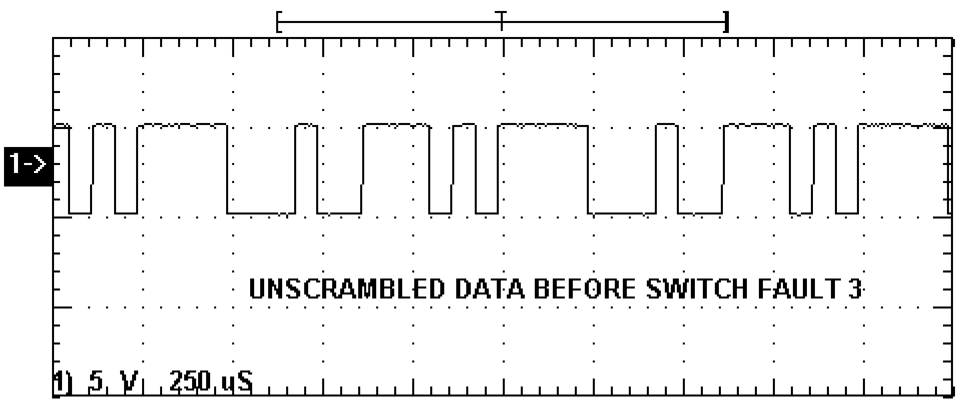
This circuit is used for unscrambling the scrambled data. After the process of unscrambling the original input data is recovered. The scrambled data is shifted using U6 and U7 (IC74HCT74) a D- flip-flop. The D-flip-flop is clocked using an external clock. The shifted data and the original data are EX-ORed. Hence the scrambled data is unscrambled and the original data is recovered as per the input data.

PROCEDURE

First keep switch **3** of **SF1** to OFF position.

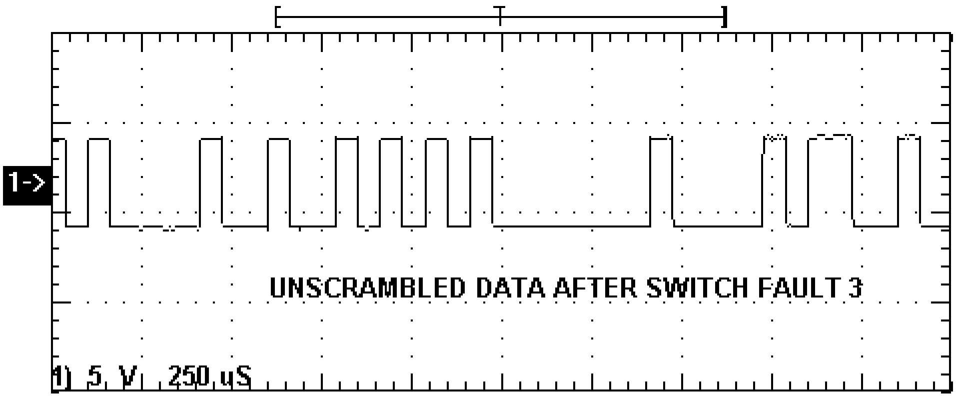
1. Connect power supply in proper polarity to the kit **DCL-BBT** and switch it ON.
2. Keep the switch **SW1** as per the position shown in the block diagram fig.6.1 to generate the PRBS data pattern.
3. Connect the **CLK OUT** post of the PRBS generator to the **CLK IN** post of the Scrambler.
4. Connect the **DATA OUT** post to the **DATA IN** post of the Scrambler.
5. Connect the scrambled data from the **OUT** post of the Scrambler to the **DATA IN** post of the Un-scrambler.
6. Connect the **CLK OUT** post of the PRBS generator to the **CLK IN** post of the Un-scrambler.
7. Observe the output data at the **OUT** post of the Un-scrambler.

Waveform before Switch Fault



Now put the switch **3** of **SF1** in the Switch Fault to ON position and observe its effect.

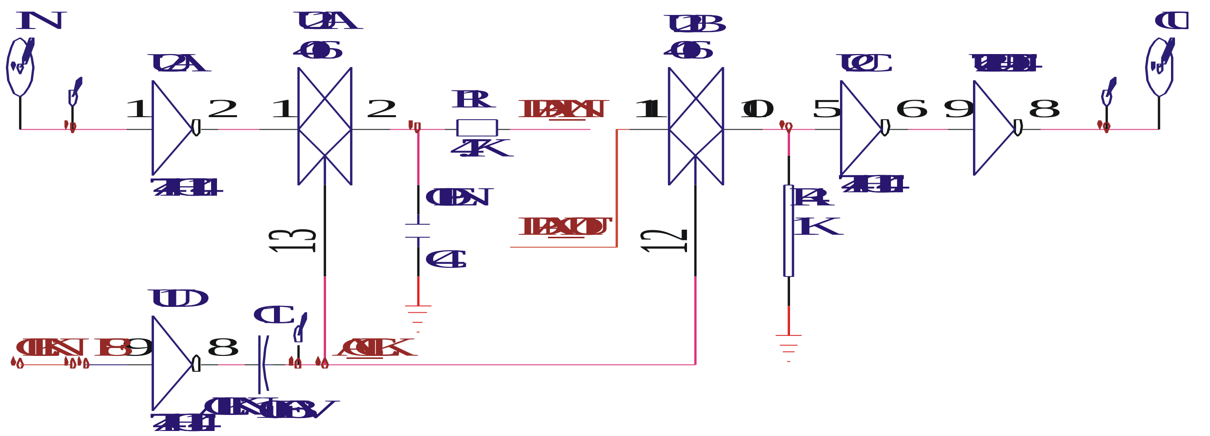
Waveform after Switch Fault



Action and Effect

When the switch **3** of **SF1** is put to ON position, the clock going to pin no.11 of U6 gets opened due to which the synchronization is mismatched. There is loss in the recovered data.

4) Effect of Switch Fault 4 (DCL-BBT) in PAM section



DESCRIPTION

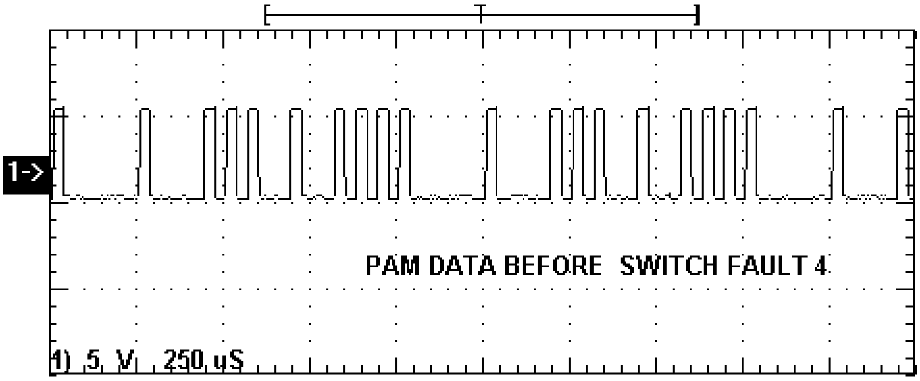
This section provides the pulse modulated signal output at the PAM OUT Post. The Pulse Amplitude Modulation circuitry is built around the analog switch U21 (CD4016). The amplitudes of the regularly spaced rectangular pulses vary in direct proportion to the instantaneous sample values of continuous digital pulses.

PROCEDURE

First keep switch **4** of **SF1** to OFF position.

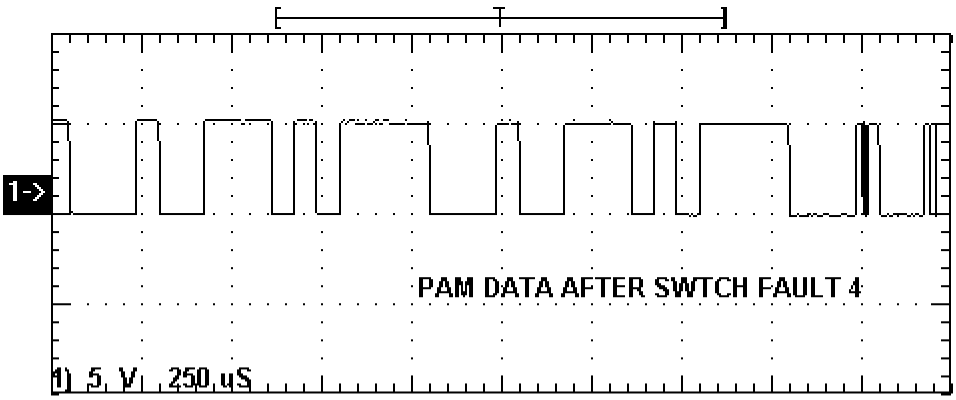
1. Connect power supply in proper polarity to the kits **DCL-BBT** and switch it ON.
2. Keep the switch of the PRBS generator **SW1** as shown in the Fig.1.1 for the generation of the PRBS pattern.
3. Set the clock frequency of the PRBS generator to minimum i.e. **16 KHz** using jumper **JP1**.
4. Connect the PRBS data from the **DATA OUT** post of PRBS generator to the **IN** post of **PAM**.
5. Connect channel 1 of the oscilloscope to the **DATA OUT** post of the PRBS generator and channel 2 to the **OUT** post of the **PAM** section.
6. Observe the pulse amplitude modulated digital data on the oscilloscope.

Waveform before Switch Fault



Now put the switch **4** of **SF1** in the Switch Fault to ON position and observe its effect.

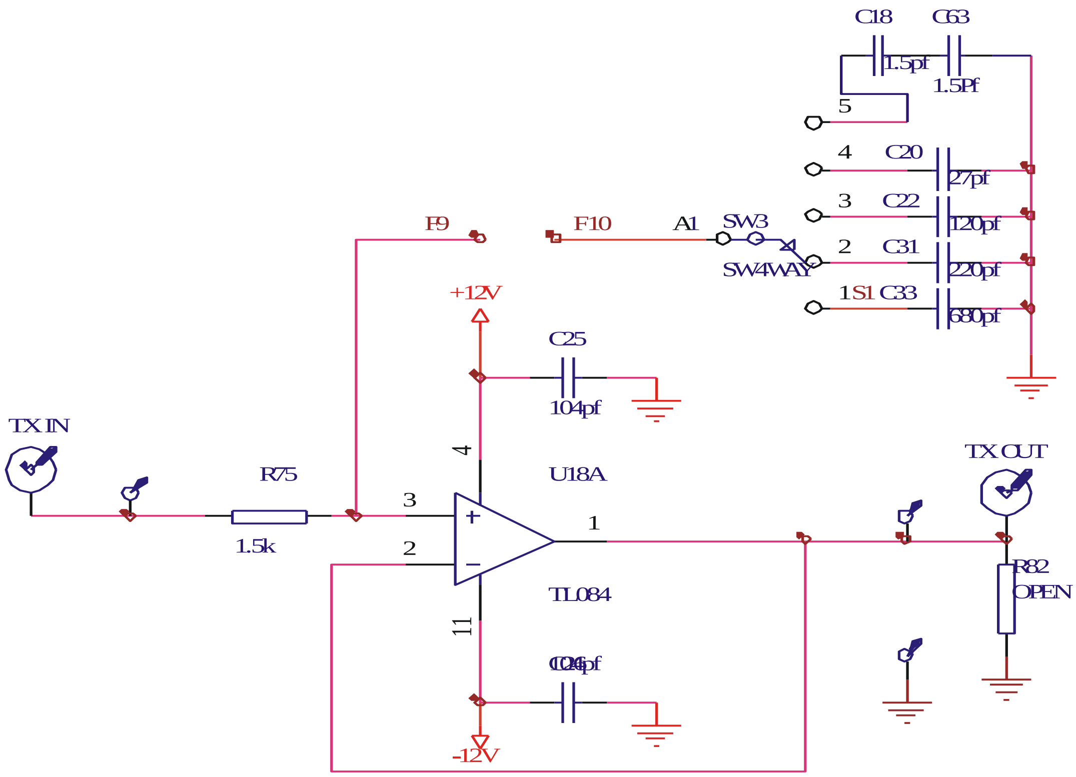
Waveform after Switch Fault



Action and Effect

When the switch **4** of **SF1** is put to ON position, the clock going to pin no.9 of U1 gets disconnected due to which the modulation process is disturbed.

5) Effect of Switch Fault 5 (DCL-BBT) in TX filter section



DESCRIPTION

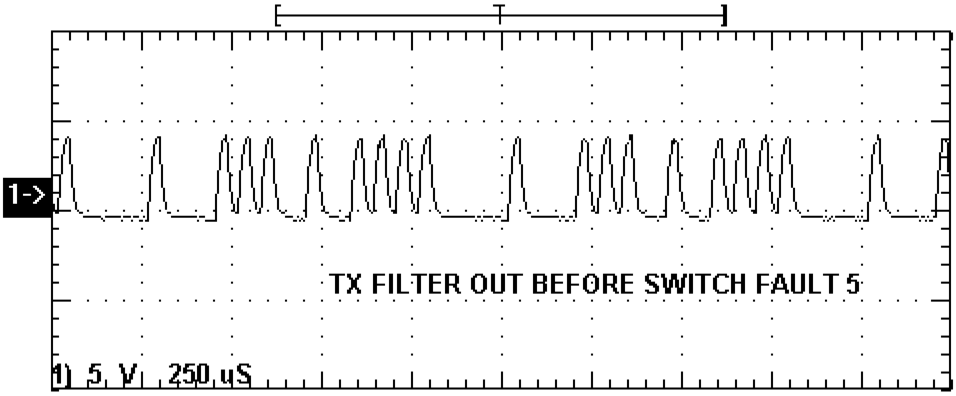
The transmitter filter consists of a first order low pass filter built around U18 (ICTL084) .The bandwidth of the filter is varied by selecting capacitors of different values (C18, C20, C22, C31, C33) using rotary switch SW3.

PROCEDURE

First keep switch **5** of **SF2** to OFF position.

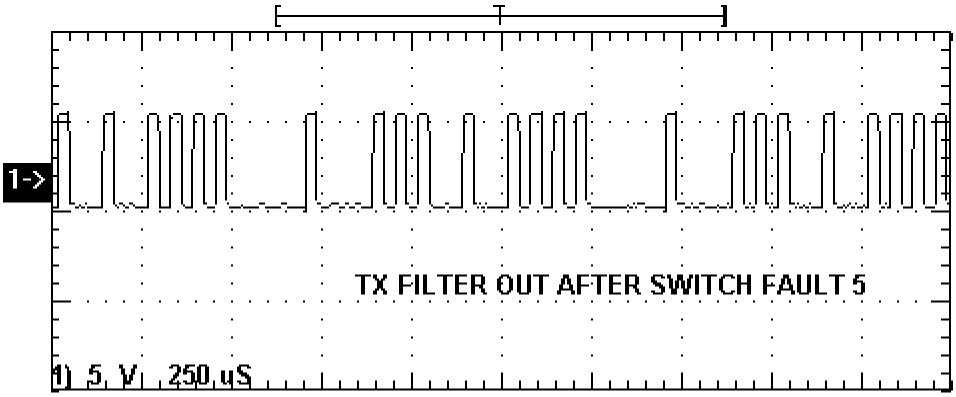
1. Connect power supply in proper polarity to the kits **DCL-BBT** and switch it ON.
2. Keep the switch of the PRBS generator **SW1** as shown in the Fig.2.1 for the generation of the PRBS pattern.
3. Select the **PRBS** clock frequency to **16 KHz** using jumper **JP1**.
4. Connect the PRBS **DATA OUT** post to **DATA IN** post of the **PAM.**
5. Keep the position of the rotary switch **SW2** of TX channel to **1**.
6. Connect the PAM **OUT** post to the **TX IN** of the TX Filter**.**
7. Observe the filter response at the **TX OUT** post of the TX filter.

Waveform before Switch Fault



Now put the switch **5** of **SF2** in the Switch Fault to ON position and observe its effect.

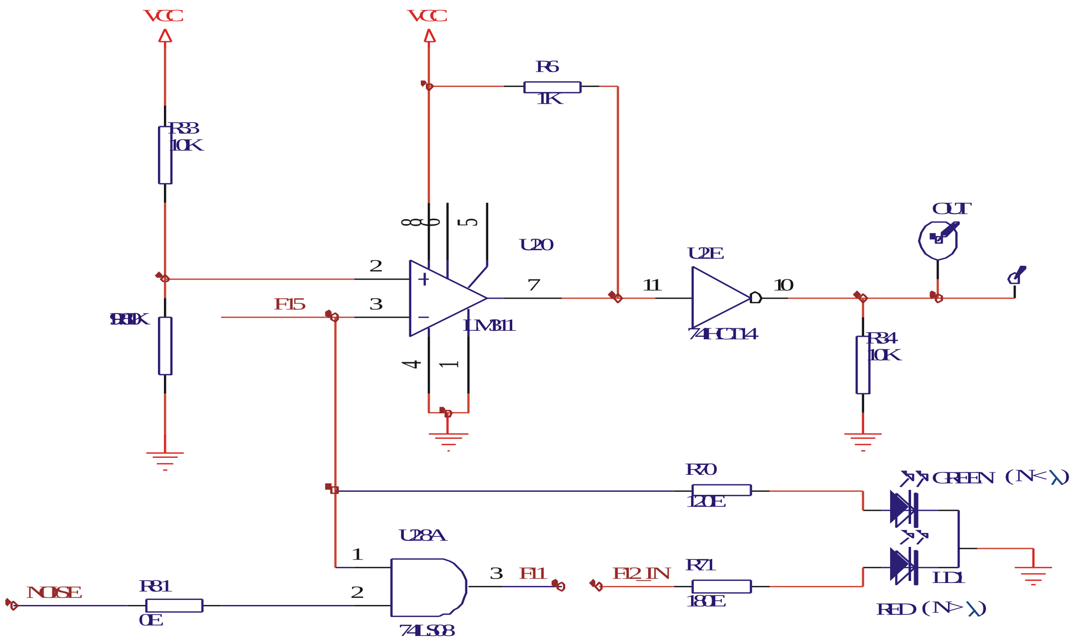
Waveform after Switch Fault



Action and Effect

When the switch **5** of **SF2** is put to ON position, the connection going to the rotary switch **SW3** which in turn is connected to the capacitor gets opened due to which there is no proper filtration. Hence there will be no effect of bandwidth in the transmitted data.

6) Effect of Switch Fault 6 (DCL-BBT) in decision device section



DESCRIPTION

The decision device consists of a comparator circuit built around U20 (IC LM311). A threshold voltage is fixed using a voltage divider circuit. The detected output of the decision device is buffered using U2 (IC74HCT14) .The bi-color LED is used to detect the fine data and the noisy data with two different color indications namely green and red.

PROCEDURE

First keep switch **6** of **SF2** to OFF position.

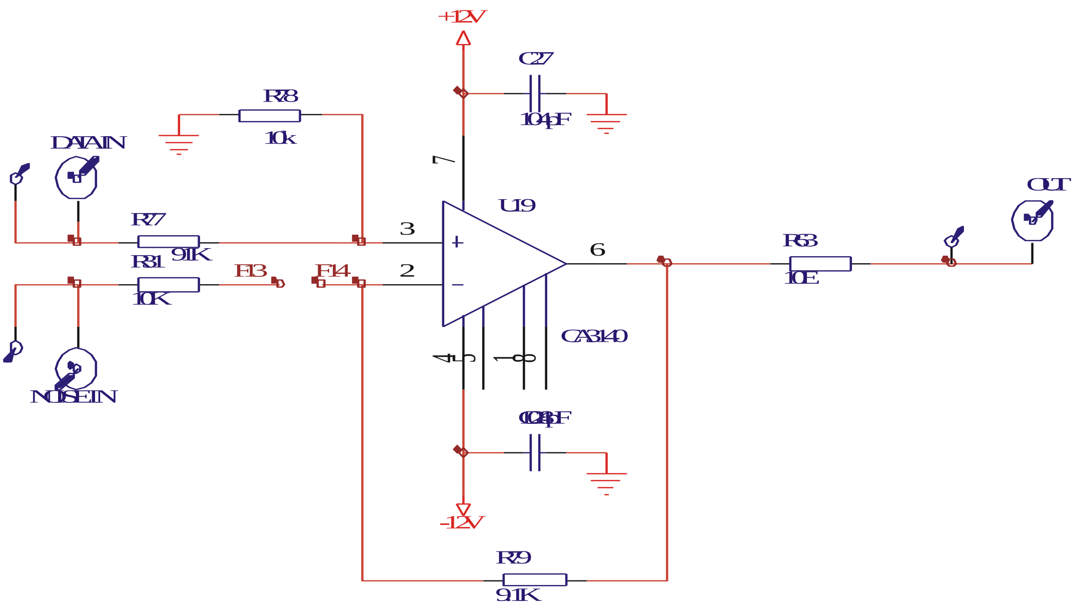
1. Connect power supply in proper polarity to the kits **DCL-BBT** and switch it ON.
2. Keep the switch of the PRBS generator **SW1** as shown in the Fig.2.1 for the generation of the PRBS pattern.
3. Select the **PRBS** clock frequency to **16 KHz** using jumper **JP1**.
4. Connect the PRBS **DATA OUT** post to **DATA IN** post of the **PAM.**
5. Keep the position of the rotary switch **SW2** of TX channel to **1**.
6. Connect the PAM **OUT** post to the **TX IN** of the TX Filter and connect the data from the **TX OUT** post to the **DATA IN** post of the Adder.
7. Keep the noise level to the minimum position by rotating pot **P1** anti clockwise.
8. Connect the noise from the **NOISE OUT** post of the noise source to the **NOISE IN** post of the Adder.
9. Connect the **OUT** post of the Adder to the **RX IN** post of the RX filter.
10. Keep the position of the rotary switch **SW3** of the RX filter to **1**.
11. Connect the data from the **RX OUT** post to the **IN** post of the Decision device.
12. Note that **GREEN** LED will glow if noise level is below threshold level and **RED** LED will glow when noise level increases more than the threshold level.
13. Connect the **OUT** post of the Decision device to the **DATA IN** post of the Regenerator.
14. Connect the **CLK OUT** from the PRBS generator to the **CLK IN** post of the Regenerator.

15. Observe the output at the **OUT** post of the Regenerator for various data format.
16. Now slowly increase the level of noise from the noise generator. Observe the output. Note that red LED glows indicating the presence of noise in the transmitted data. Hence no data is derived at the output.
17. Now put the switch **6** of **SF2** in the Switch Fault to ON position and observe its effect.

Action and Effect

When the switch **6** of **SF2** is put to ON position, the connection of pin no.3 of U28 going to one of the lead of the noise indicator LED is opened. Hence the noise indicator LED with red color indication does not respond even when the noise level is high in the system.

7) Effect of Switch Fault 7 (DCL-BBT) In The Adder Section



DESCRIPTION

The Adder circuit adds the noise and the data and gives a noisy data at the output of the adder, whenever the noise level is high. Adder circuit is built around U19 (IC CA3140).

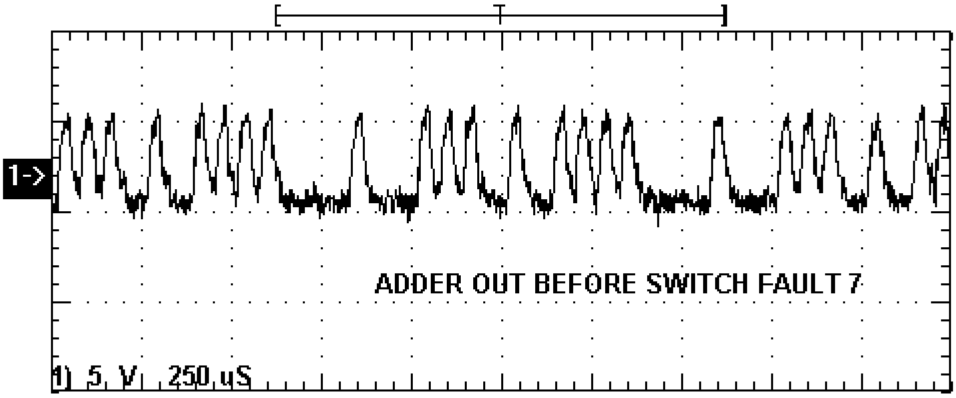
PROCEDURE

First keep switch **7** of **SF2** to OFF position.

1. Connect power supply in proper polarity to the kits **DCL-BBT** and switch it ON.
2. Keep the switch of the PRBS generator **SW1** as shown in the Fig.2.1 for the generation of the PRBS pattern.
3. Select the **PRBS** clock frequency to **16 KHz** using jumper **JP1**.
4. Connect the PRBS **DATA OUT** post to **DATA IN** post of the **PAM.**
5. Keep the position of the rotary switch **SW2** of TX channel to **1**.
6. Connect the PAM **OUT** post to the **TX IN** of the TX Filter and connect the data from the **TX OUT** post to the **DATA IN** post of the Adder.

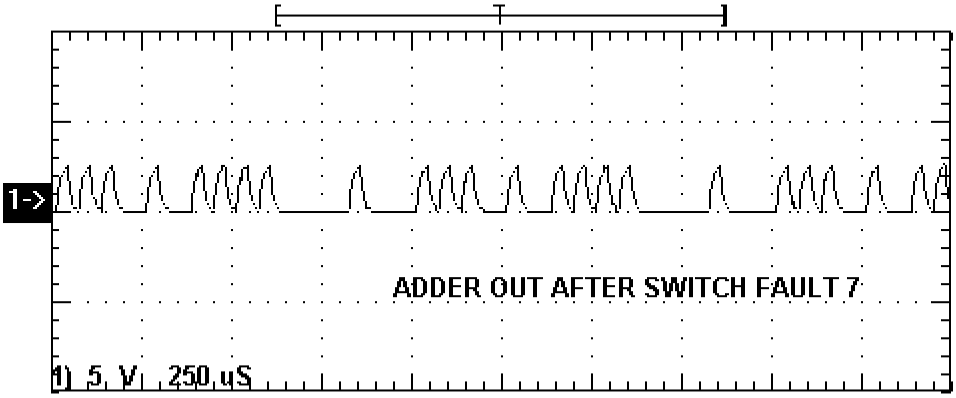
7. Keep the noise level to the minimum position by rotating pot **P1** anti clockwise.
8. Connect the noise from the **NOISE OUT** post of the noise source to the **NOISE IN** post of the Adder.
9. Increase the noise to the maximum level by rotating pot **P1** in a clockwise direction. Observe the waveform at the **OUT** post of the Adder.

Waveform before Switch Fault



Now put the switch **7** of **SF2** in the Switch Fault to ON position and observe its effect.

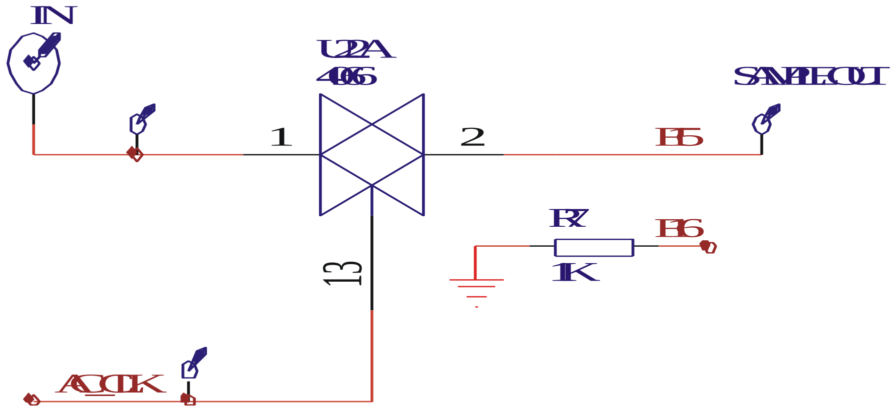
Waveform after Switch Fault



Action and Effect

When the switch **7** of **SF2** is put to ON position, the noise in going to pin no.2 of gets opened due to which the noise does not gets added in the data. It also affects the entire circuitry function of the Adder.

8) Effect of Switch Fault 8 (DCL-BBT) in the sampler of the receiver section



DESCRIPTION

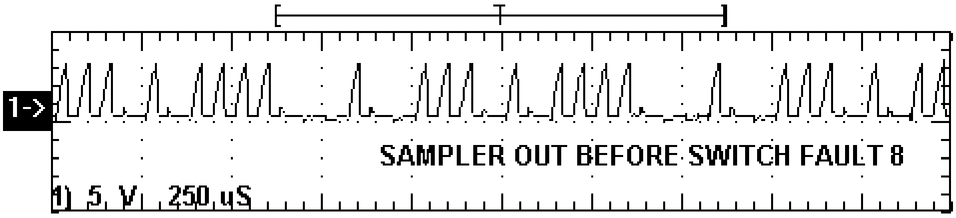
This circuit samples the received data at various clock frequencies. The sampling frequency depends upon the clock position we have selected. The sampler circuit is built around U22 (4066).

PROCEDURE

First keep switch **8** of **SF2** to OFF position.

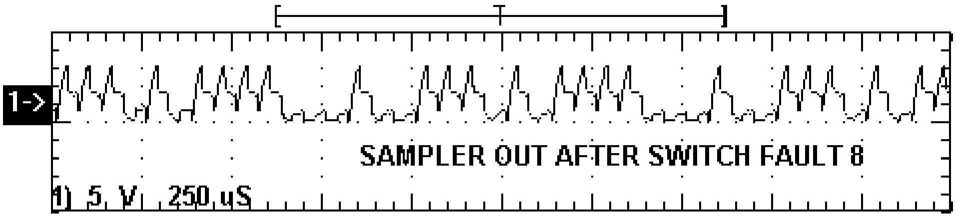
1. Connect power supply in proper polarity to the kits **DCL-BBT** and switch it ON.
2. Keep the switch of the PRBS generator **SW1** as shown in the Fig.2.1 for the generation of the PRBS pattern.
3. Select the **PRBS** clock frequency to **16 KHz** using jumper **JP1**.
4. Connect the PRBS **DATA OUT** post to **DATA IN** post of the **PAM.**
5. Keep the position of the rotary switch **SW2** of TX channel to **1**.
6. Connect the PAM **OUT** post to the **TX IN** of the TX Filter and connect the data from the **TX OUT** post to the **DATA IN** post of the Adder.
7. Keep the noise level to the minimum position by rotating pot **P1** anti clockwise.
8. Connect the noise from the **NOISE OUT** post of the noise source to the **NOISE IN** post of the Adder.
9. Connect the **OUT** post of the Adder to the **RX IN** post of the RX filter.
10. Keep the position of the rotary switch **SW3** of the RX filter to **1**.
11. Connect the data from the **RX OUT** post to the **IN** post of the Decision device.
12. Observe the output at the output test point of the sampler in the Decision device section.

Waveform before Switch Fault



Now put the switch **8** of **SF2** in the Switch Fault to ON position and observe its effect.

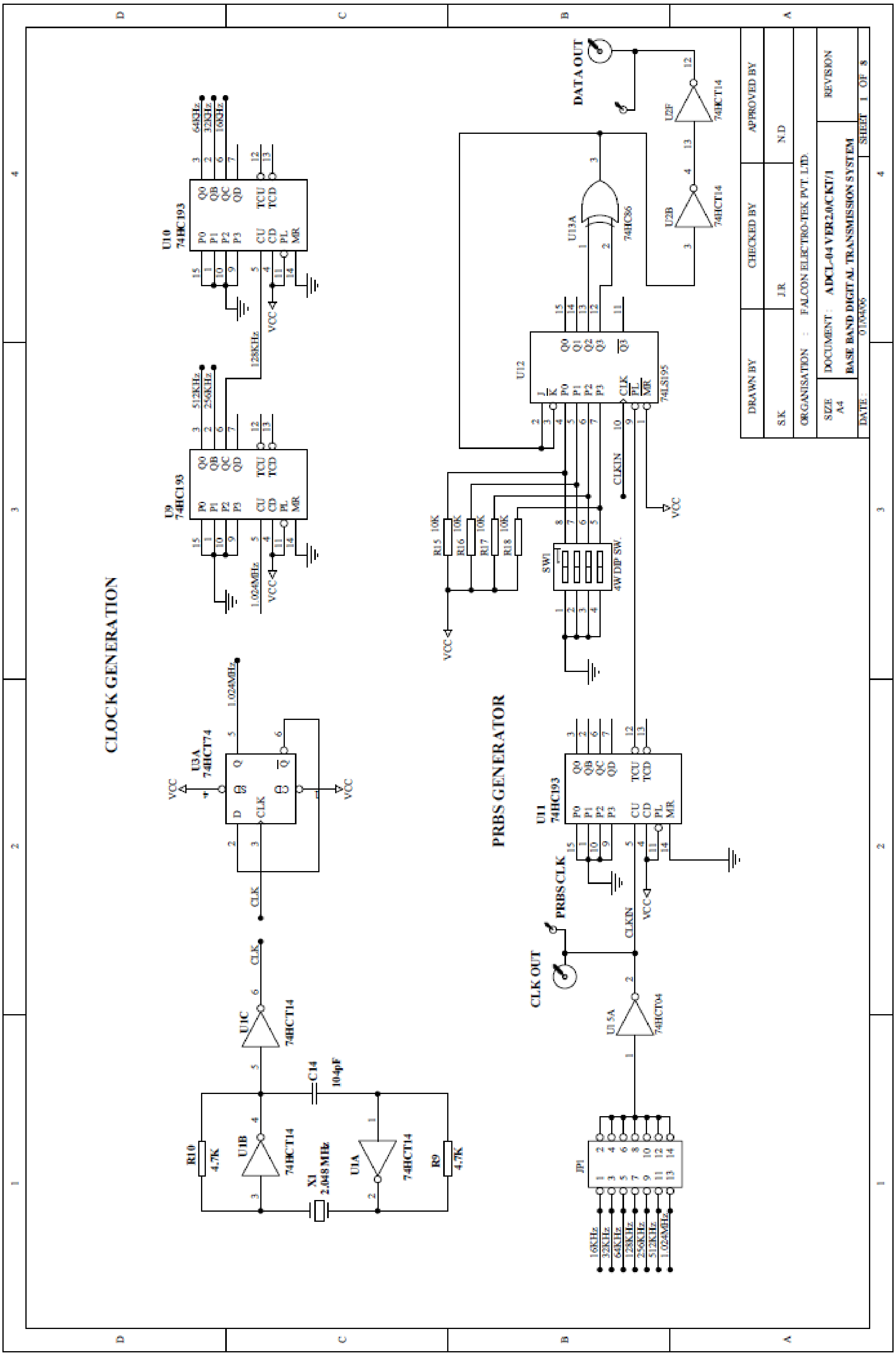
Waveform after Switch Fault

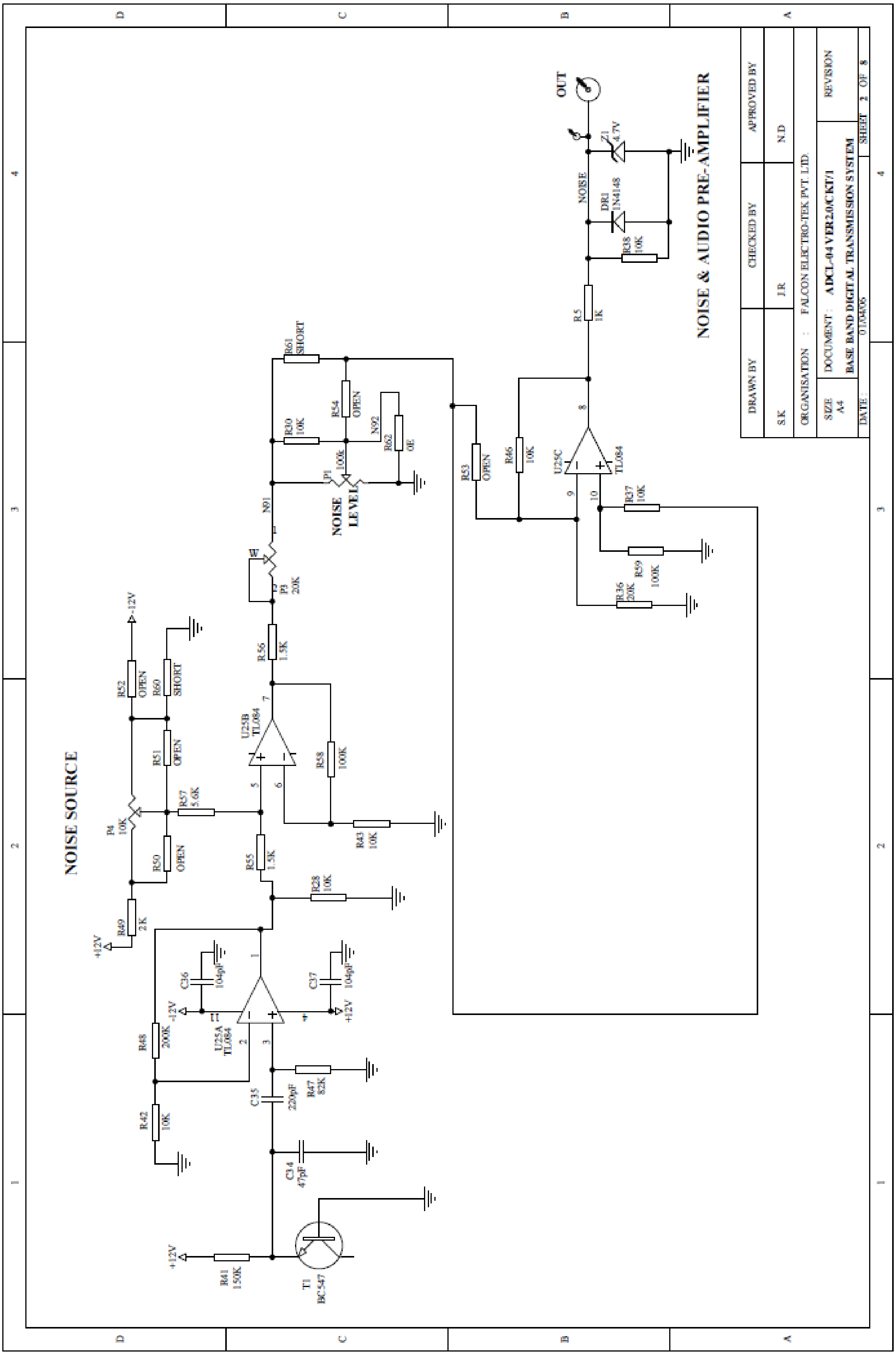


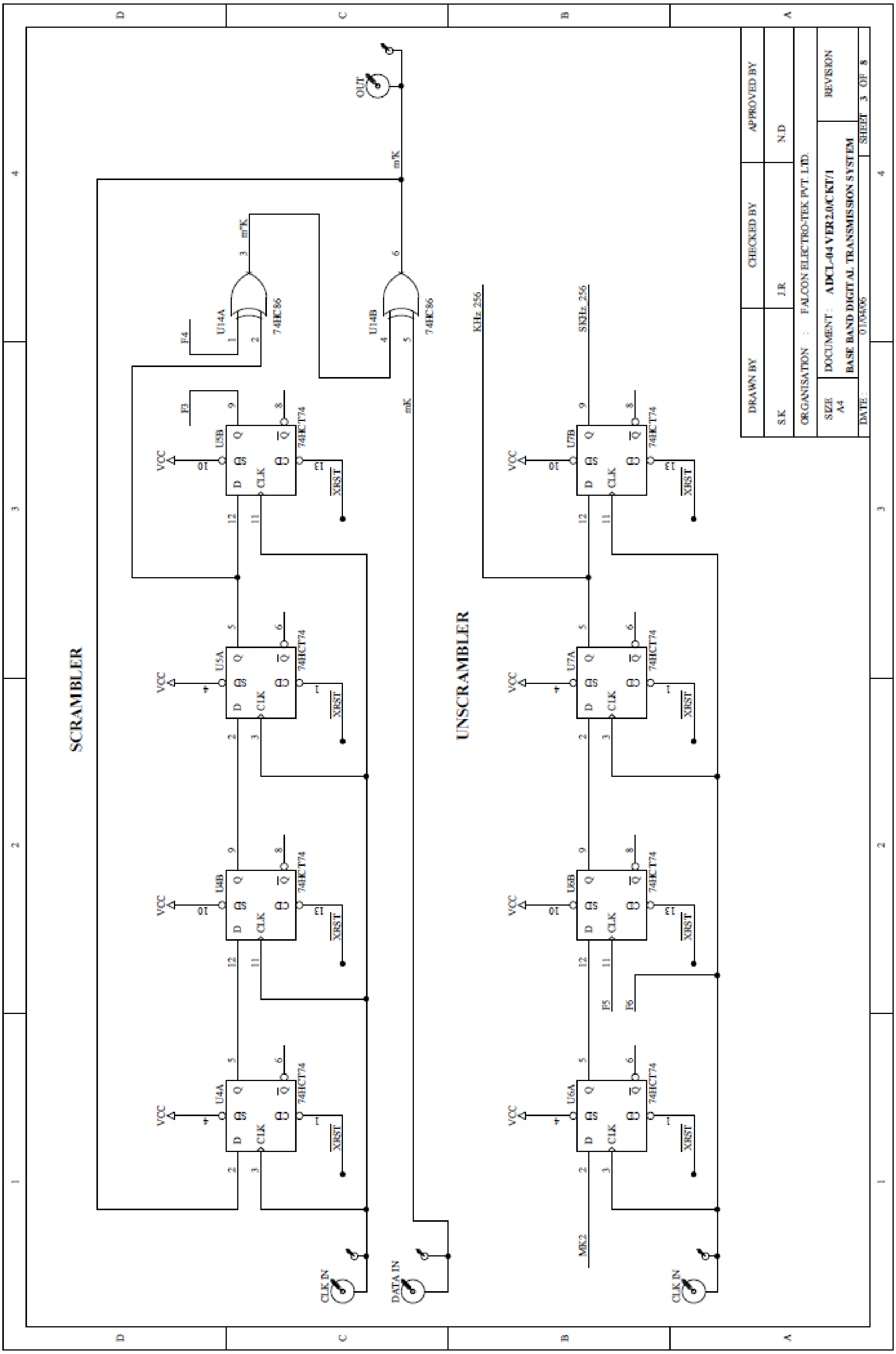
Action and Effect

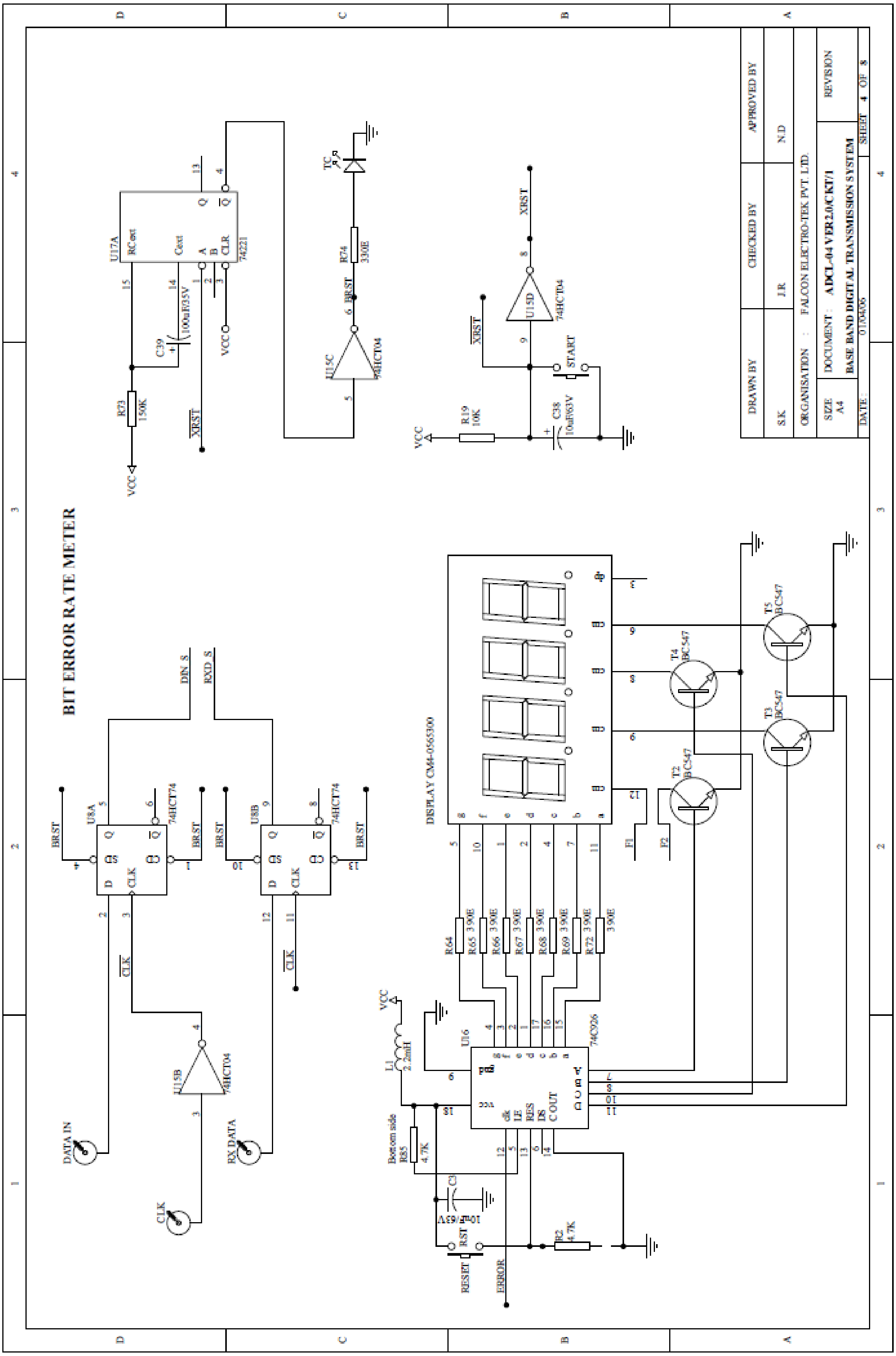
When the switch **8** of **SF2** is put to ON position the load resistor R7 gets opened due to which the shape of the sampled output gets affected.

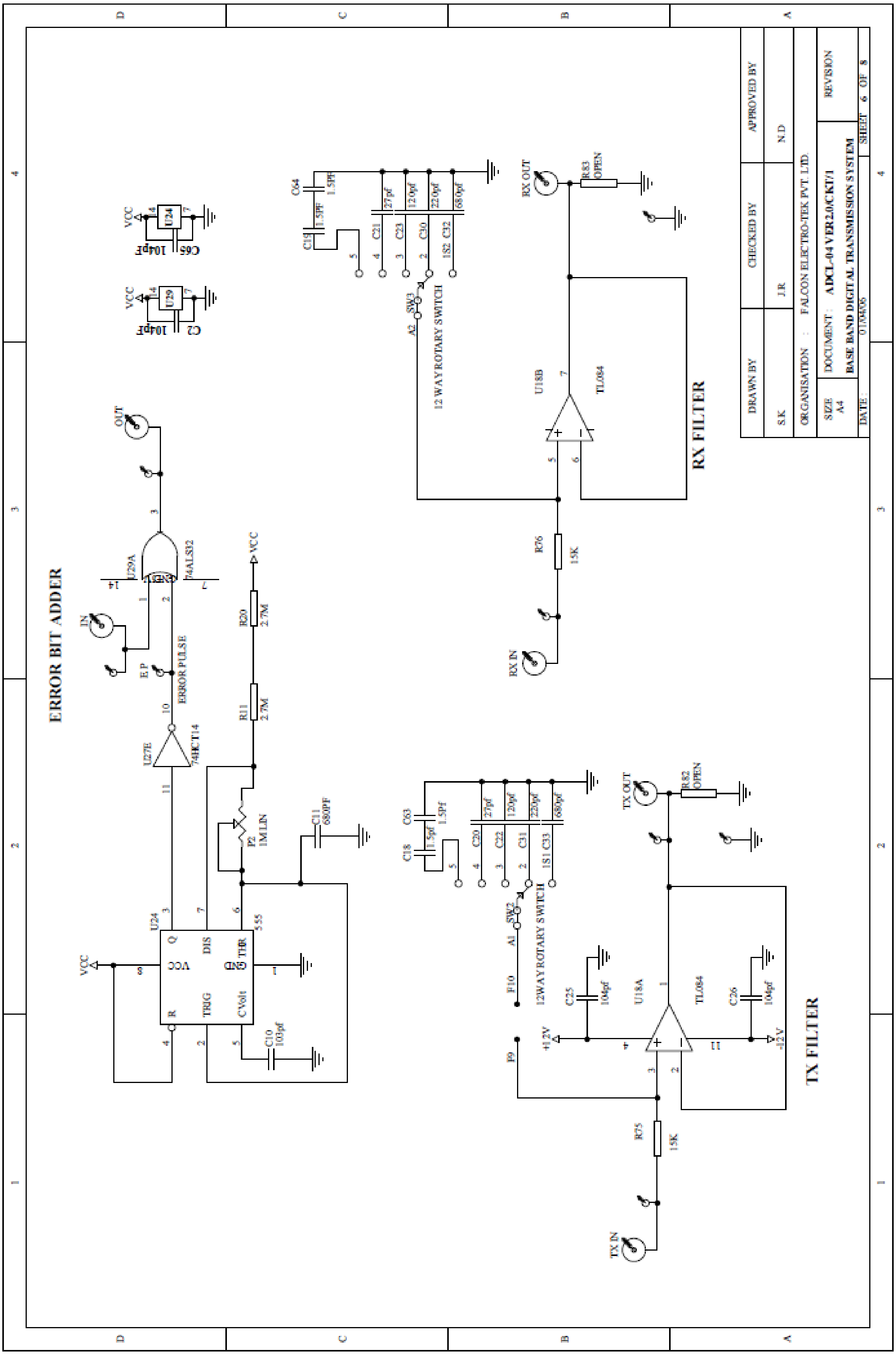
CIRCUIT DIAGRAM

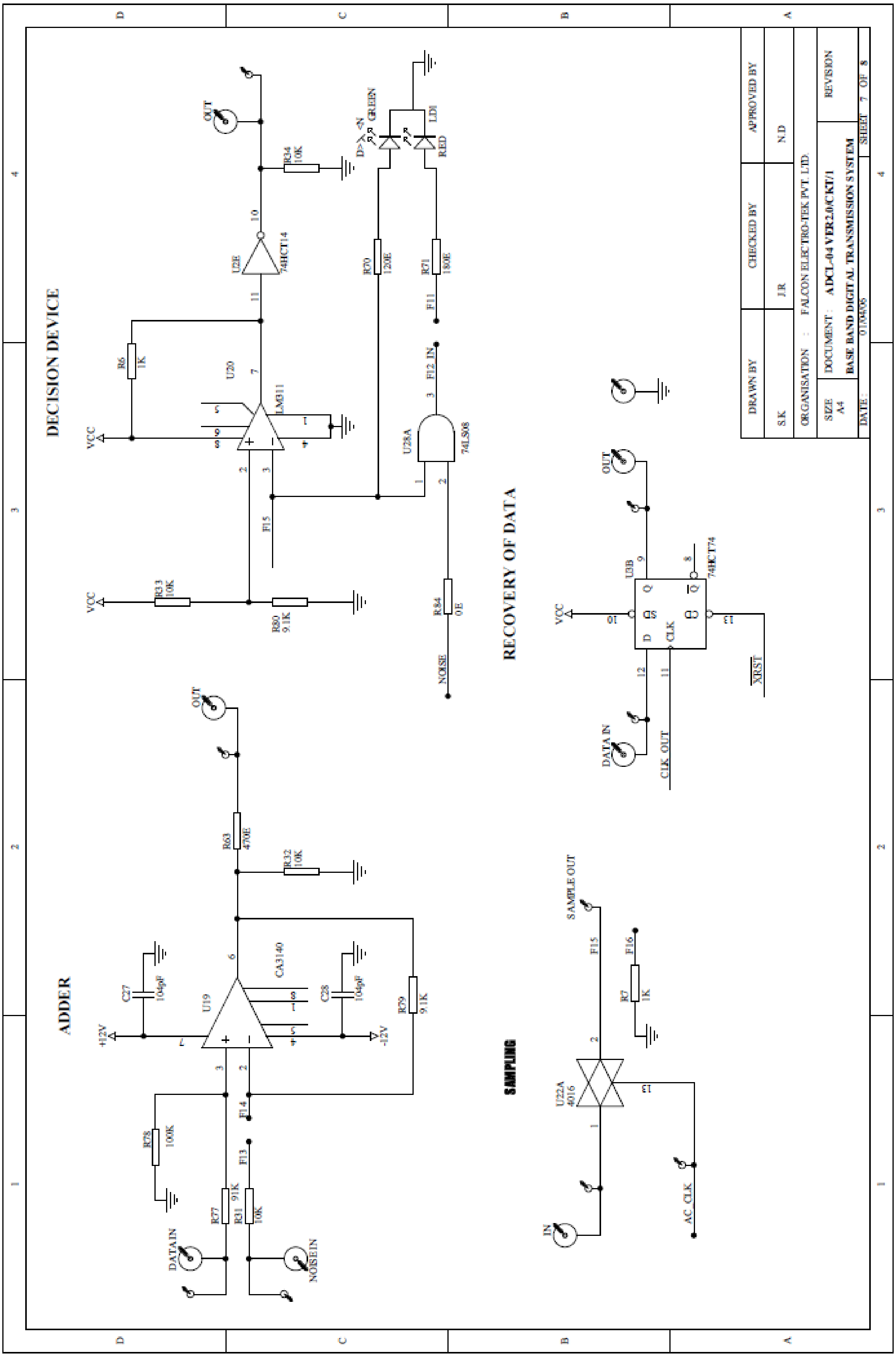


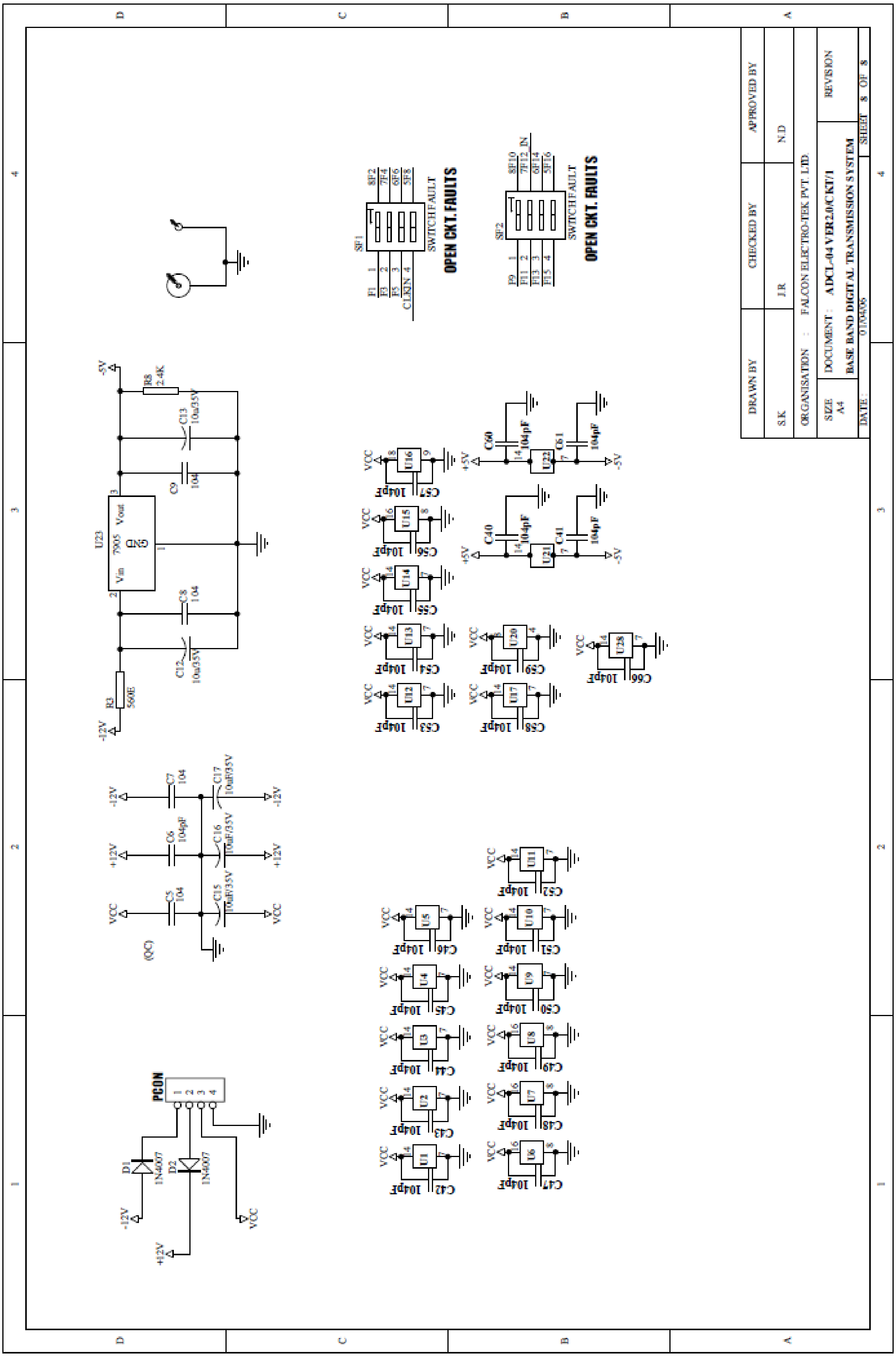

NOISE SOURCE
NOISE & AUDIO PRE-AMPLIFIER
NOISE LEVEL
OUT
NOISE
DRAWN BY
S.K
CHECKED BY
J.R
APPROVED BY
N.D
ORGANISATION : FALCON ELECTRO-TEK PVT. LTD.
SIZE
A4
DOCUMENT : ADCL-04 VER2.0/CKT/1
BASE BAND DIGITAL TRANSMISSION SYSTEM
REVISION
SHEET 2 OF 8

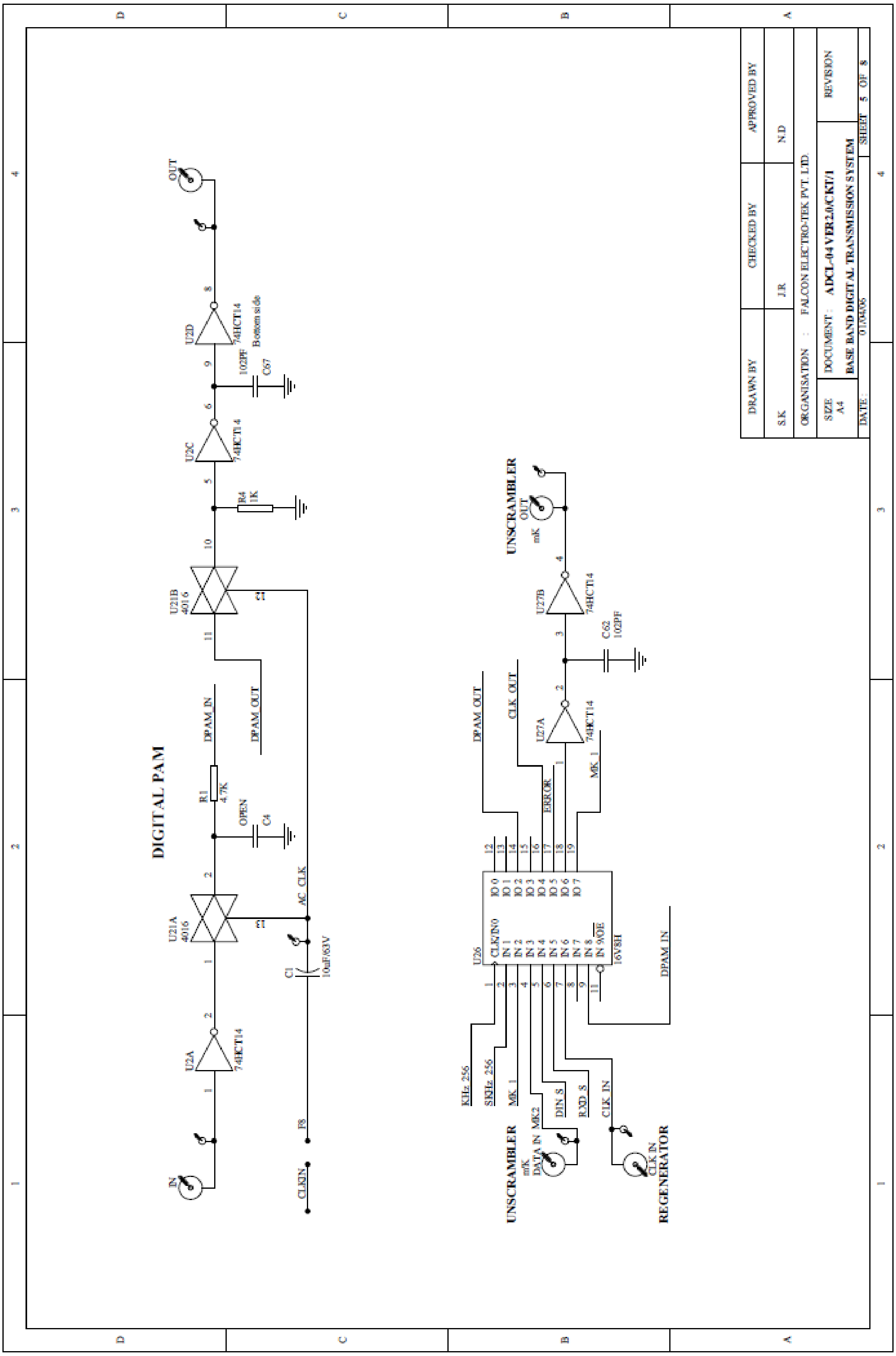

SCRAMBLER
UNSCRAMBLER
CLK IN
DATA IN
OUT
CLK IN
U4A
U4B
U5A
U5B
U6A
U6B
U7A
U7B
U14A
U14B
74HCT74
74HC86
VCC
XRST
mK
m'K
MK2
DRAWN BY
S.K
CHECKED BY
J.R
APPROVED BY
N.D
ORGANISATION : FALCON ELECTRO-TEK PVT LTD.
DOCUMENT : ADCL-04 VER2.0/CKT/1
BASE BAND DIGITAL TRANSMISSION SYSTEM
SIZE A4
REVISION
SHEET 3 OF 8
DATE


BIT ERROR RATE METER
FALCON ELECTRO-TEK PVT. LTD.
BASE BAND DIGITAL TRANSMISSION SYSTEM


ERROR BIT ADDER
RX FILTER
TX FILTER
12 WAY ROTARY SWITCH
TL084
74ALS32
74HCT14
555
ERROR PULSE
RX IN
RX OUT
TX IN
TX OUT
FALCON ELECTRO-TEK PVT. LTD.
ADCL-04 VER2.0/CKT/1
BASE BAND DIGITAL TRANSMISSION SYSTEM
SHEET 6 OF 8


DECISION DEVICE
ADDER
RECOVERY OF DATA
SAMPLING
DRAWN BY
CHECKED BY
APPROVED BY
S.K
J.R
N.D
ORGANISATION : FALCON ELECTRO-TEK PVT. LTD.
SIZE A4
DOCUMENT : ADCL-04 VER2.0/CKT/1
BASE BAND DIGITAL TRANSMISSION SYSTEM
REVISION
DATE : 01/04/06
SHEET 7 OF 8


OPEN CKT. FAULTS
SWITCH FAULT
PCON
DRAWN BY
CHECKED BY
APPROVED BY
ORGANISATION : FALCON ELECTRO-TEK PVT. LTD.
DOCUMENT : ADCL-04 VER2.0/CKT/1
BASE BAND DIGITAL TRANSMISSION SYSTEM
REVISION
SIZE A4
SHEET 8 OF 8


DIGITAL PAM
IN
OUT
U2A
74HCT14
U21A
4016
U21B
4016
U2C
74HCT14
U2D
74HCT14
Bottom side
R1
4.7K
R4
1K
C4
OPEN
C67
102PF
C1
10uF/63V
DPAM IN
DPAM OUT
AC CLK
CLKIN
P8
UNSCRAMBLER
UNSCRAMBLER
REGENERATOR
U26
16V8H
U27A
74HCT14
U27B
74HCT14
C62
102PF
DPAM OUT
CLK OUT
ERROR
MK 1
KHz 256
SKHz 256
DIN S
RXD S
CLK IN
DATA IN
MK2
DPAM IN
DRAWN BY
S.K
CHECKED BY
J.R
APPROVED BY
N.D
ORGANISATION : FALCON ELECTRO-TEK PVT. LTD.
SIZE
A4
DOCUMENT : ADCL-04 VER 2.0/CKT/1
BASE BAND DIGITAL TRANSMISSION SYSTEM
REVISION
DATE : 01/04/06
SHEET 5 OF 8

EXPERIMENT NO. 1

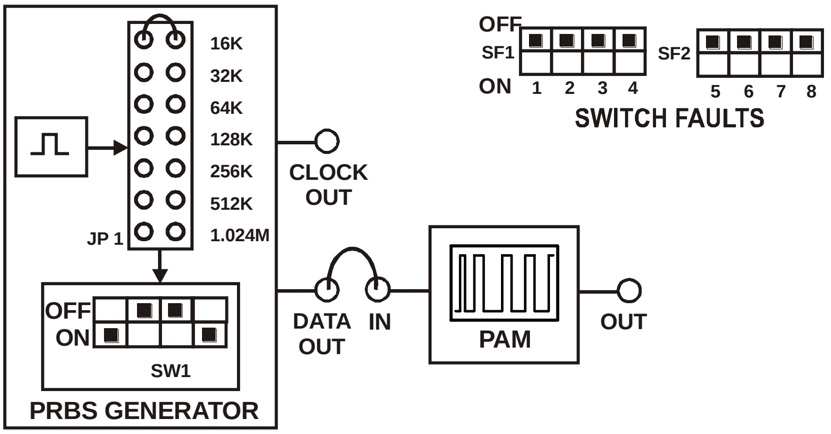


FIG. 1.1 BLOCK DIAGRAM FOR STUDY OF PULSE AMPLITUDE MODULATION OF DIGITAL DATA FOR BASE BAND TRANSMISSION

EXPERIMENT NO: 1

NAME

Study of Pulse Amplitude Modulation of Digital Data for Digital Baseband Transmission

OBJECTIVE

To study the pulse amplitude modulation of digital data for baseband digital transmission

THEORY

Digital message representation in baseband commonly takes the form of an amplitude modulated pulse train. We express such signals by writing

$$X(t) = \sum_{k} a_k p(t - kD) \qquad (1)$$

Where the modulating amplitude a_k represents the kth symbol in the message sequence, so the amplitude belong to a set of M discrete values. The index k ranges from $-\infty$ to $+\infty$ unless otherwise stated. Equation (1) defines a digital PAM signal, as distinguished from those rare cases when pulse duration or pulse position modulation is used for digital transmission.

The un-modulated pulse $p(t)$ may be rectangular or some other shape, subject to the conditions

$$p(t) = \begin{cases} 1 & t = 0 \\ 0 & t = \pm D, \pm 2D, \ldots \end{cases} \qquad (2)$$

This condition ensures that we can recover the message by sampling x (t) periodically at t = KD, $K = 0, \pm 1, \pm 2\ldots$ since

$$X(KD) = \sum_{k} a_k p(KD - kD) = a_{k]}$$

In the use of discrete pulse modulation, where the amplitude, duration, or the position of the transmitted pulses is varied in a discrete manner in accordance with the given data stream. The use of discrete pulse amplitude modulation for the baseband transmission of digital data is one of the most efficient schemes in terms of power and bandwidth utilization.

EQUIPMENTS

- DCL-BBT kit
- Connecting Chords
- Power supply
- 20MHz Dual Trace Oscilloscope

NOTE: Keep The Switch Faults In Off Position.

PROCEDURE

1. Refer to the block diagram (Fig.1.1) and carry out the following connections and switch settings.
2. Connect power supply in proper polarity to the kits **DCL-BBT** and switch it ON.
3. Keep the switch of the PRBS generator **SW1** as shown in the Fig.1.1 for the generation of the PRBS pattern.
4. Set the clock frequency of the PRBS generator to minimum i.e. **16 KHz** using jumper **JP1**.
5. Connect the PRBS data from the **DATA OUT** post of PRBS generator to the **IN** post of **PAM**.
6. Connect channel 1 of the oscilloscope to the **DATA OUT** post of the PRBS generator and channel 2 to the **OUT** post of the **PAM** section.
7. Observe the pulse amplitude modulated digital data on the oscilloscope.
8. Vary the clock frequencies of the PRBS data from 16 KHz to 1.024 MHz with the help of jumper **JP1** provided on board and observe the pulse amplitude modulated data at various clock frequencies.

OBSERVATION

Observe the PRBS data at **DATA OUT** at various clock frequencies and PAM output at **OUT**.

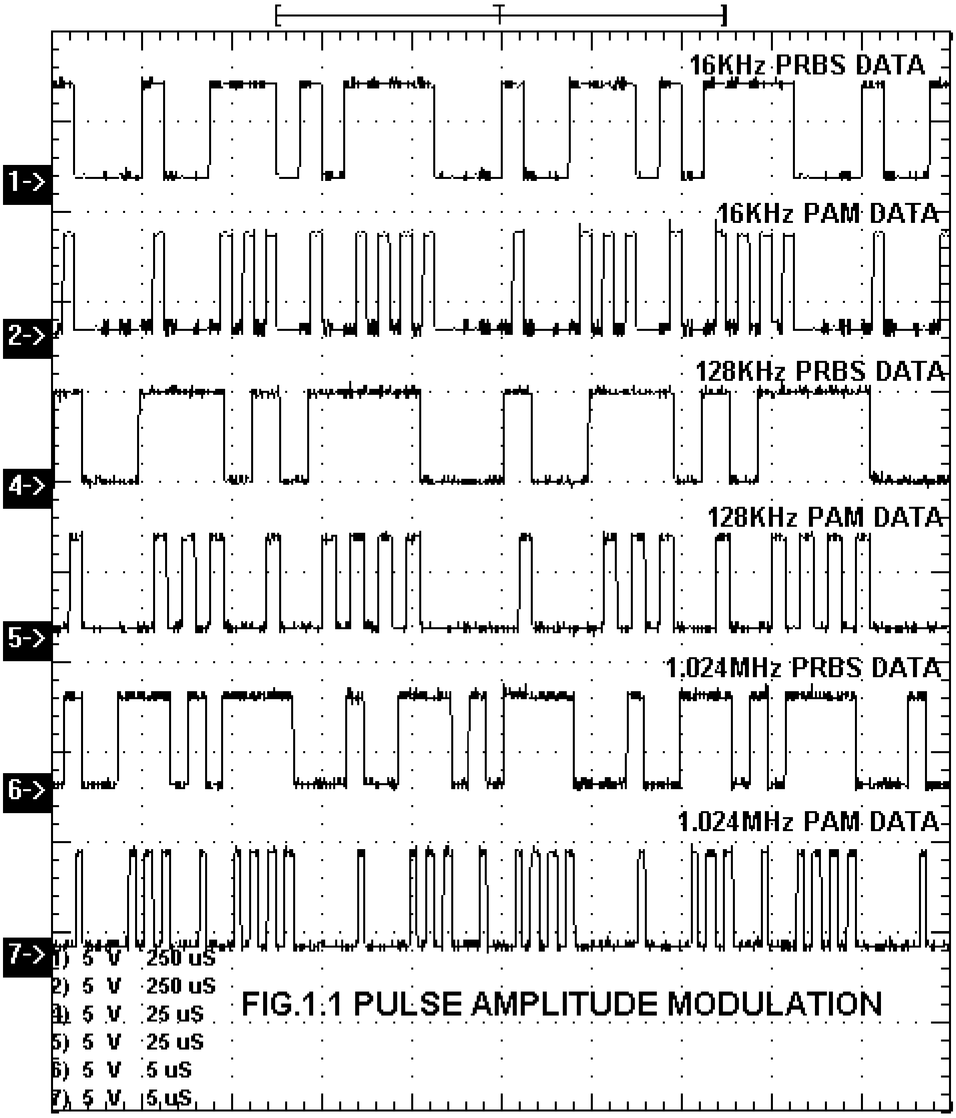


FIG.1.1 PULSE AMPLITUDE MODULATION

EXPERIMENT NO. 2

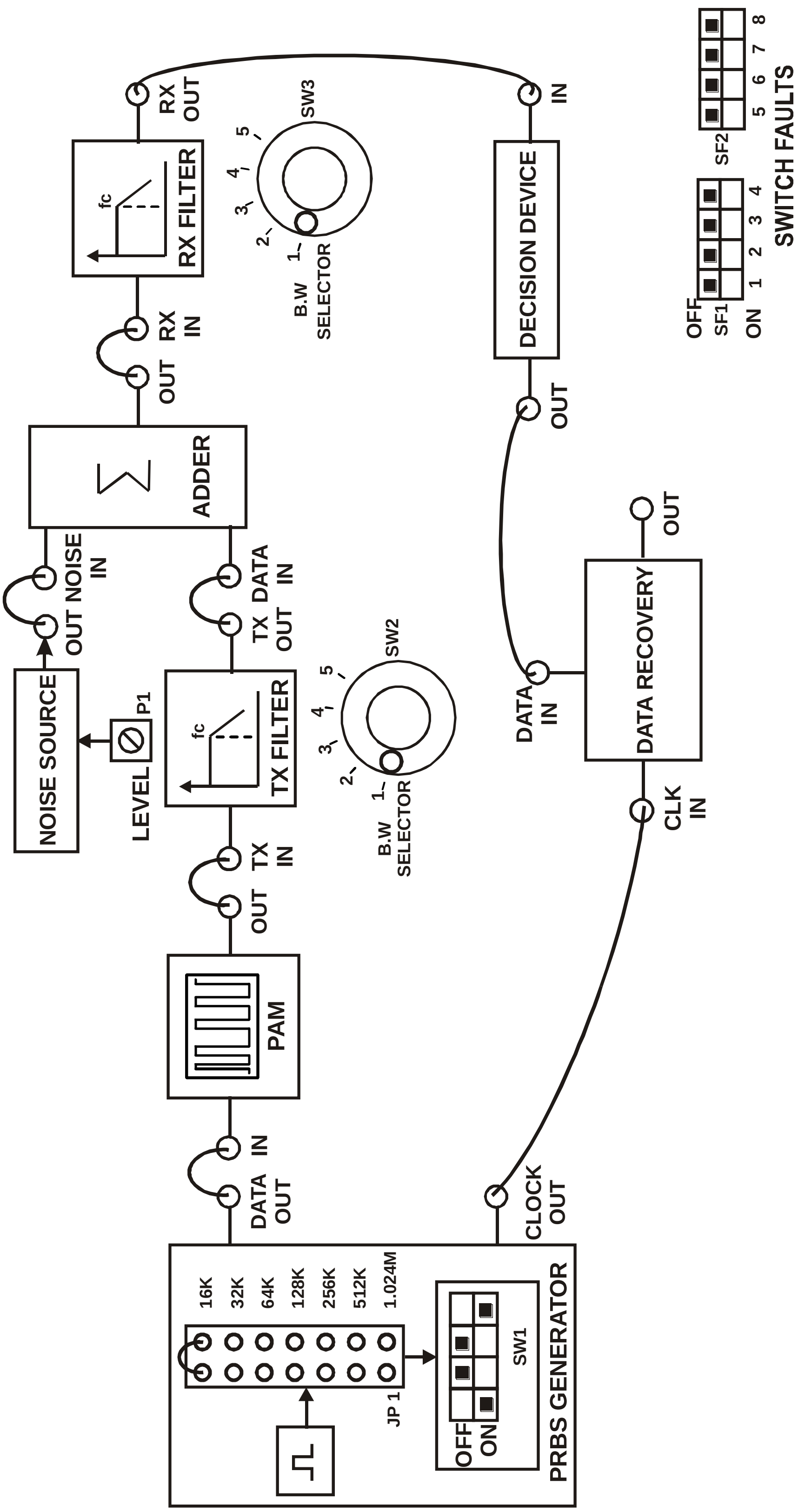


FIG. 2.1 BLOCK DIAGRAM FOR STUDY OF EXTRACTION AND RECOVERY OF DATA IN BASE BAND DIGITAL TRANSMISSION

EXPERIMENT NO: 2

NAME

Study of Extraction and Recovery of Data in Baseband Digital Transmission

OBJECTIVE

To study the baseband digital transmission system

THEORY

For the baseband transmission of digital data, the use of discrete pulse amplitude modulation (PAM) is one of the most efficient schemes in terms of power and bandwidth utilization.

In accordance with the block diagram the incoming binary sequence (b_k) consists of symbols 1 & 0, each of duration T_b. The pulse amplitude modulator modifies the binary sequence (b_k) into a new series of short pulses, whose amplitude a_k is represented in a polar form

$$a_k = \begin{cases} +1 & \text{if symbol } b_k \text{ is } 1 \\ -1 & \text{if symbol } b_k \text{ is } 0 \end{cases}$$

The sequence of short pulses so produced is applied to a transmit filter of impulse response g (t), producing the transmitted signal

$$S(t) = \sum_k a_k g(t - kT_b)$$

The signal s (t) is modified as a result of transmission through the channel of impulse response h (t).

In addition, the channel adds random noise to the signal at the receiver input. The noisy signal x (t) is then passed through a receiver filter of impulse response c (t). The resulting filter output y (t) is sampled synchronously with the transmitter, with the sampling instants being determined by a clock or timing signal. Finally the sequence of samples thus obtained is used to reconstruct the original data sequence by means of a decision device. Specifically, the amplitude of each sample is compared to a thresholdλ. If the threshold λ is not exceeded decision is made in favor of symbol 0.If the sample amplitude equals to the threshold the device will determine which symbol was transmitted.

EQUIPMENTS

- DCL-BASE BAND kit
- Connecting Chords
- Power supply
- 20MHz Dual Trace Oscilloscope

NOTE: Keep The Switch Faults In Off Position**.**

PROCEDURE

1. Refer to the block diagram (Fig.2.1) and carry out the following connections and switch settings.
2. Connect power supply in proper polarity to the kits **DCL-BBT** and switch it ON.
3. Keep the switch of the PRBS generator **SW1** as shown in the Fig.2.1 for the generation of the PRBS pattern.
4. Select the **PRBS** clock frequency to **16 KHz** using jumper **JP1**.
5. Connect the PRBS **DATA OUT** post to **DATA IN** post of the **PAM.**
6. Keep the position of the rotary switch **SW2** of TX channel to **1**.
7. Connect the PAM **OUT** post to the **TX IN** of the TX Filter and connect the data from the **TX OUT** post to the **DATA IN** post of the Adder.
8. Keep the noise level to the minimum position by rotating pot **P1** anti clockwise.
9. Connect the noise from the **NOISE OUT** post of the noise source to the **NOISE IN** post of the Adder.
10. Connect the **OUT** post of the Adder to the **RX IN** post of the RX filter.
11. Keep the position of the rotary switch **SW3** of the RX filter to **1**.
12. Connect the data from the **RX OUT** post to the **IN** post of the Decision device.
13. Note that **GREEN** LED will glow if noise level is below threshold level and **RED** LED will glow when noise level increases more than the threshold level.
14. Connect the **OUT** post of the Decision device to the **DATA IN** post of the Regenerator.
15. Connect the **CLK OUT** from the PRBS generator to the **CLK IN** post of the Regenerator.
16. Observe the output at the **OUT** post of the Regenerator for various data format.
17. Now slowly increase the level of noise from the noise generator. Observe the output. Note that red LED glows indicating the presence of noise in the transmitted data. Hence no data is derived at the output.

OBSERVATION

Observe the following waveforms on CRO.

- TX filters **TX OUT**.
- RX filters **RX OUT**.
- Sample out on the receiver side.
- Recovered data on the Regenerator **OUT** post.

CONCLUSION

The recovered data has a one-bit delay because of the TX and TX filter response.

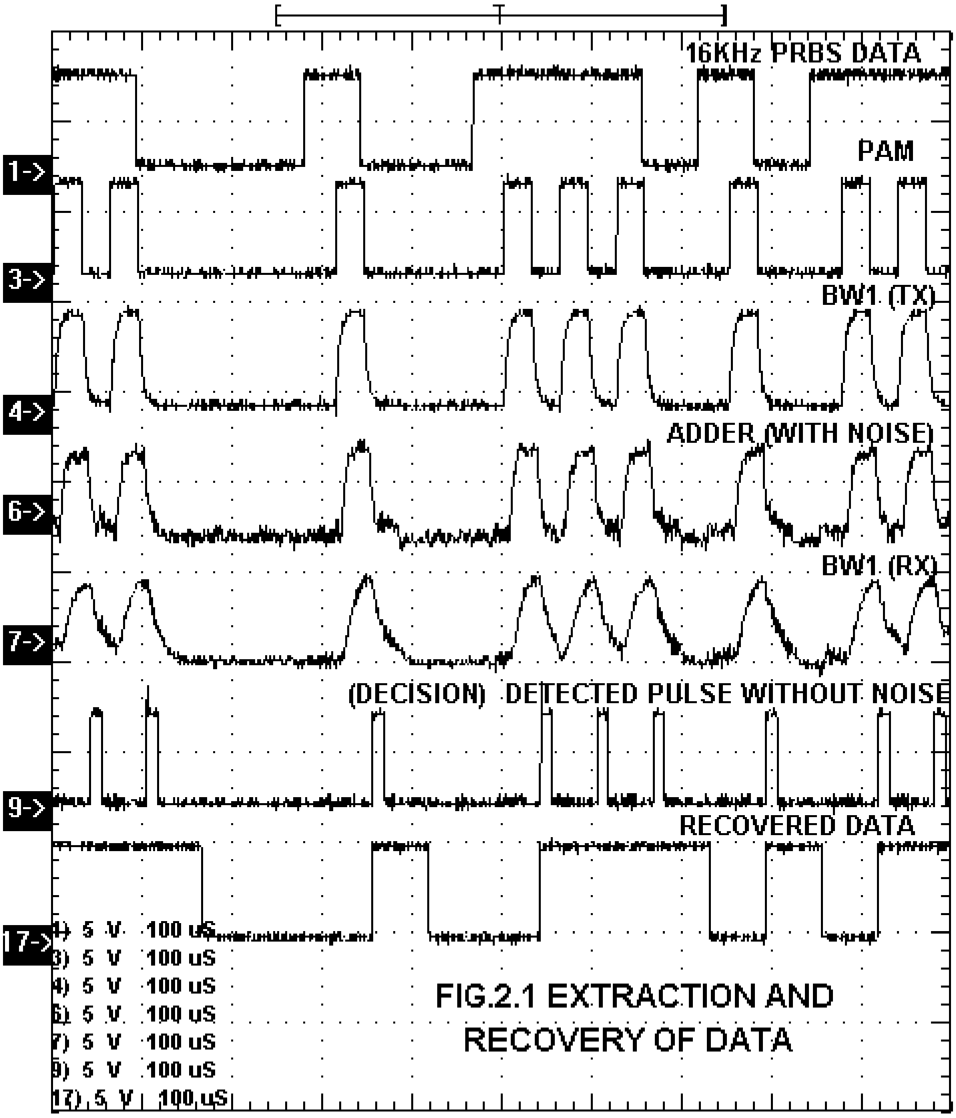


FIG.2.1 EXTRACTION AND RECOVERY OF DATA

EXPERIMENT
NO. 3

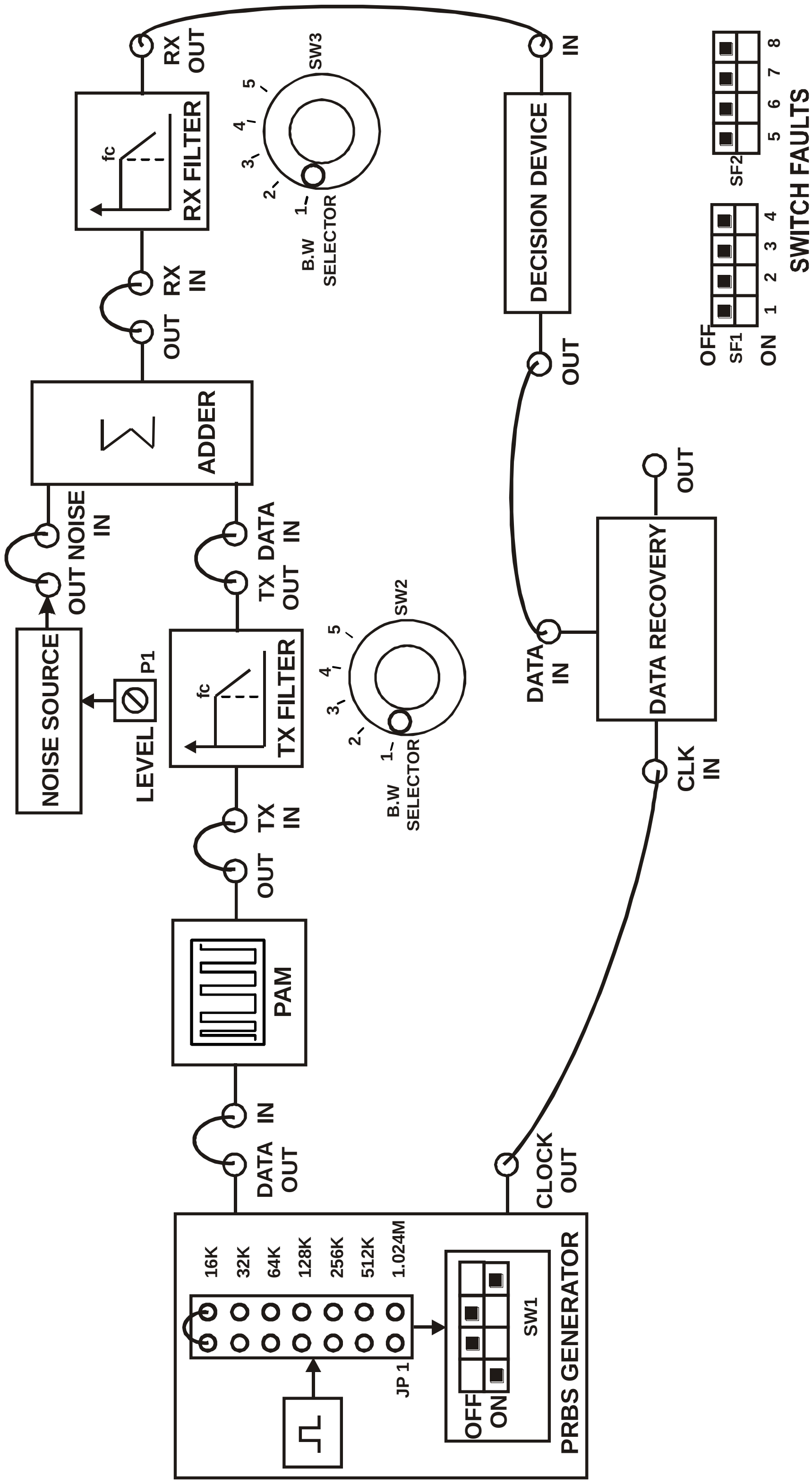


FIG. 3.1 BLOCK DIAGRAM FOR STUDY OF TRANSMISSION AND RECEPTION OF BAND LIMITED PULSE TRAIN IN BASE BAND DIGITAL TRANSMISSION SYSTEM

EXPERIMENT NO: 3

NAME

Study of Transmission and Reception of Band Limited Pulse Train in Baseband Digital Transmission System

OBJECTIVE

To study the effect of band limitation in the baseband digital transmission system

THEORY

This section develops a design for baseband digital systems when the transmission channels impose bandwidth limitation. By this we mean that the available transmission bandwidth is not large compared to the desired signaling rate and, consequently, rectangular signaling pulses would be severely distorted. Instead, we must use band-limited pulses specially shaped to avoid Inter Symbol Interference. Base-band systems therefore require the use of a low-pass channel with a bandwidth large enough to accommodate the essential frequency content of data stream and also to minimize error probabilities.

EQUIPMENTS

- DCL-BBT kit
- Connecting Chords
- Power supply
- 20MHz Dual Trace Oscilloscope

NOTE: Keep The Switch Faults In Off Position**.**

PROCEDURE

1. Refer to the block diagram (Fig.3.1) and carry out the following connections and switch settings.
2. Connect power supply in proper polarity to the kit **DCL-BBT** and switch it ON.
3. Keep the switch of the PRBS generator **SW1** as shown in the Fig.3.1 for the generation of the PRBS pattern.
4. Select the **PRBS** clock frequency to **16 KHz** using jumper **JP1**.
5. Connect the PRBS **DATA OUT** post to the **DATA IN** post of the **PAM.**
6. Keep the position of the rotary switch **SW2** of the TX channel to **1**.
7. Connect the PAM **OUT** post to the **TX IN** of the TX Filter and connect the data from the **TX OUT** post to the **DATA IN** post of the Adder.
8. Connect **TX OUT** post of TX Filter to **DATA IN** post of the Adder.
9. Connect the noise from the **NOISE OUT** post of the noise source to the **NOISE IN** post of the Adder.

10. Connect the **OUT** post of the Adder to the **RX IN** of the RX filter.
11. Keep the position of the rotary switch **SW3** of the RX filter to **1**.
12. Connect the data from the **RX OUT** to the **IN** post of the Decision device. Note that green LED will glow.
13. Connect the **OUT** post of the Decision device to the **DATA IN** post of the Regenerator.
14. Connect the **CLK OUT** from the PRBS generator to the **CLK IN** post of the Regenerator.
15. Note that the bandwidth of both the TX filter and RX filter should match.
16. Observe the output at the **OUT** post of the regenerator section on the oscilloscope.
17. Keeping the clock frequency of the PRBS data to 16 KHz change the bandwidth of the TX Filter and the RX Filter to position no.**2,** later to position **3** and position **4**.
18. Observe the filter response of both the TX and RX filters at their respective out posts at increased bandwidth.
19. Similarly transmit the PRBS data keeping the clock frequency to 256 KHz, set the bandwidth to position **5**, later to position **4** and position **3**.
20. Observe the filter response of both the TX and RX filter at their respective out posts at decreased bandwidth.
21. Observe the effect of the bandwidth on the output data at the **OUT** post of the Regenerator.

OBSERVATION

Observe the band limited pulses at the **TX OUT** post and **RX OUT** post on the oscilloscope. For different frequencies of PRBS data pattern set the bandwidth as follows:

Clock frequency	Bandwidth position
1. 16 KHz	Position 1
2. 32 KHz	Position 2
3. 64 KHz	Position 3
4. 128 KHz	Position 4
5. 256 KHz	Position 5

Mismatch the bandwidth settings of TX filter and RX filter and observe the variations in the output.

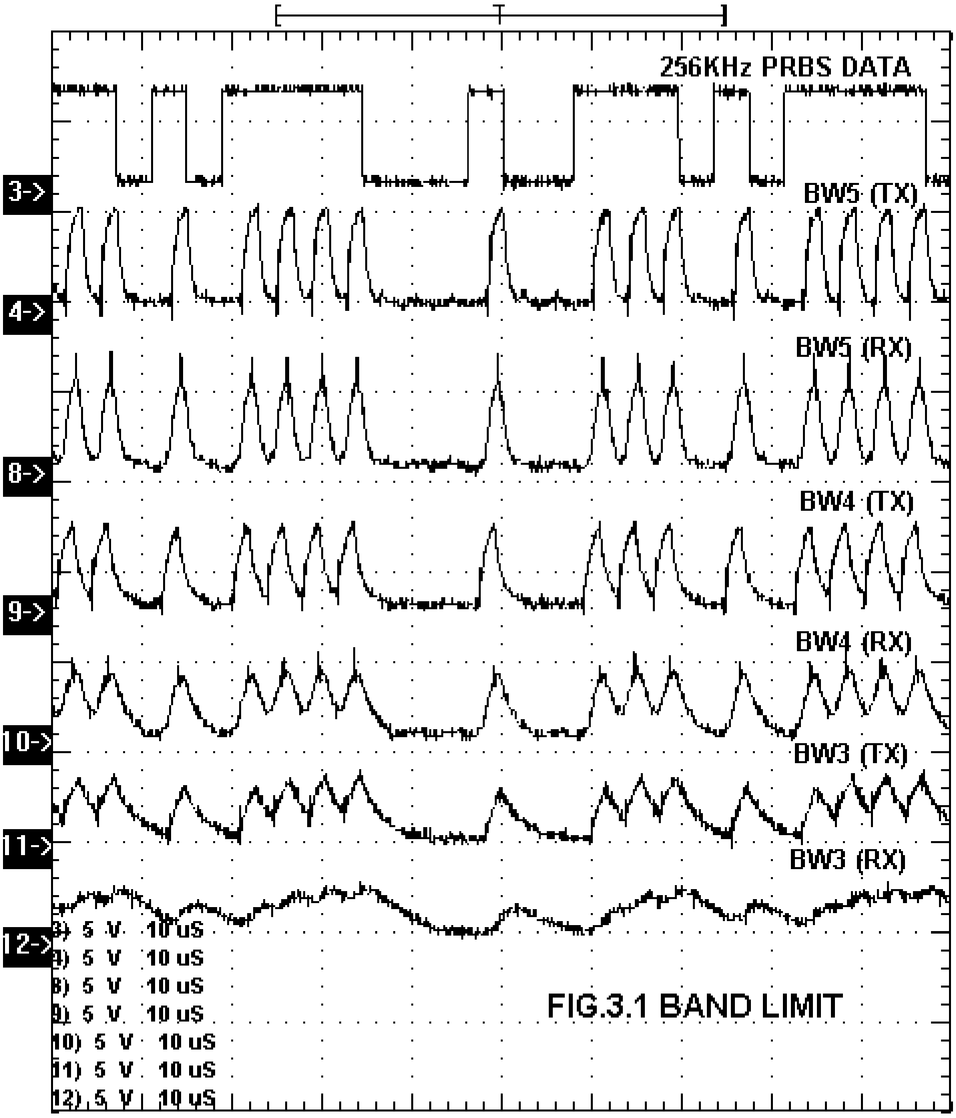


FIG.3.1 BAND LIMIT

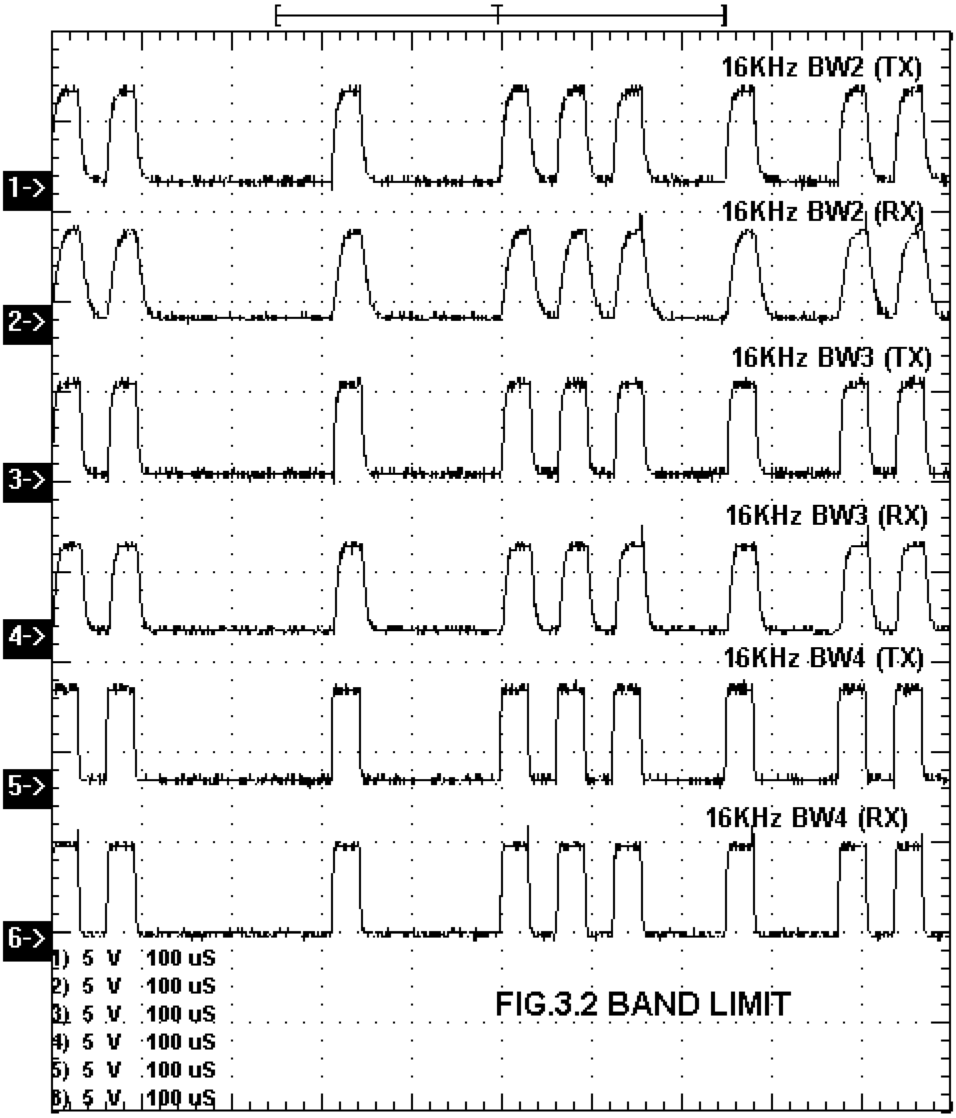


FIG.3.2 BAND LIMIT

EXPERIMENT NO. 4

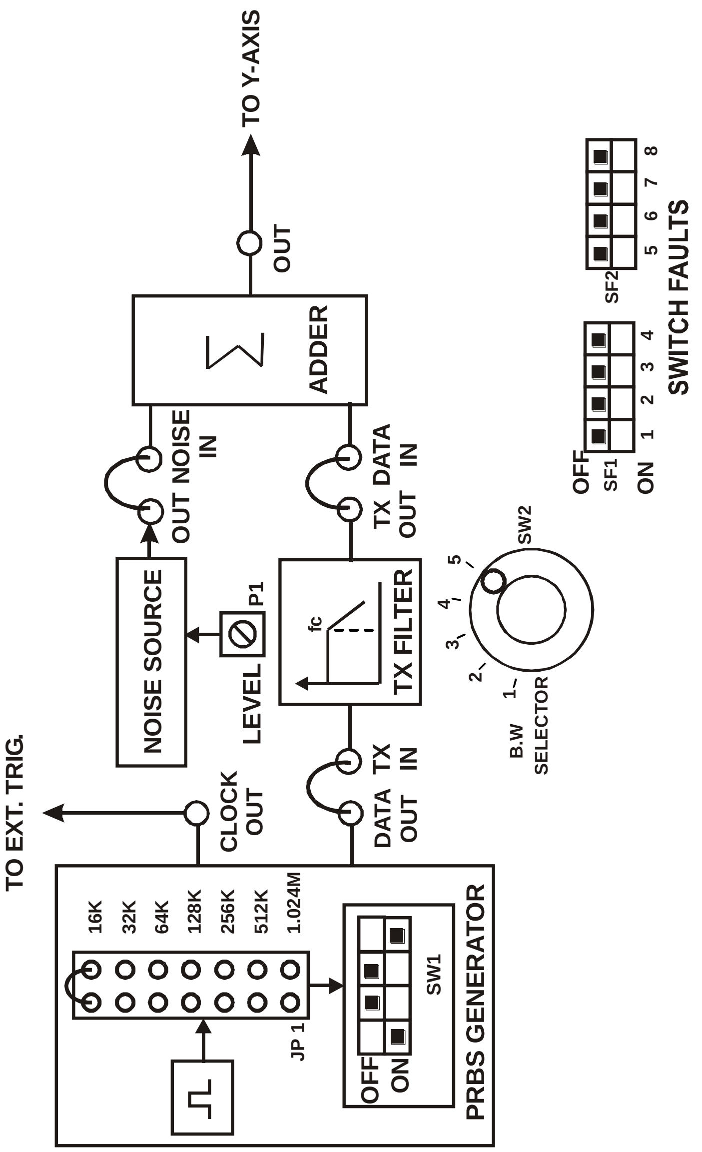


FIG. 4.1 BLOCK DIAGRAM FOR STUDY OF EYE PATTERN

EXPERIMENT NO: 4

NAME

Study of Eye Pattern

OBJECTIVE

- To study the eye pattern.
- Observation and calculation of noise margin percentage.
- Observation and calculation of timing jitter.
- Observation of the effect of Inter Symbol Interference in eye pattern.

THEORY

The eye pattern technique is a simple but powerful measurement method for accessing the data-handling ability of a digital transmission system. This method has been used extensively for evaluating the performance of wire systems. The eye pattern measurements are made in the time domain and allow the effects of waveform distortion to be shown immediately on an oscilloscope.

An eye pattern can be observed in the process as mentioned in the Fig.5.1 the output from a pseudorandom pattern generator is applied to the vertical input of an oscilloscope and the data rate is used to rigger the horizontal sweep. This results in the type of pattern shown in the Fig.5.2, which is called the eye pattern because the display shape resembles a human eye.

To measure system performance with the eye pattern method, a variety of word pattern should be provided. A convenient approach is to generate a random data signal, because this is the characteristic of data streams found in practice. This type of signal generates ones and zeros at a uniform rate but in a random manner. A variety of pseudorandom pattern generators are available for this purpose. The word pseudorandom means that the generated combination or sequence of ones and zeros will eventually repeat but that is sufficiently random for test purposes. A pseudorandom bit sequence comprises four different 2-bit-long combinations, eight different 3-bit-long combinations, sixteen different 4-bit-long combinations, and so on (i.e. sequences of different N-bit-long combinations) up to a limit set by the instrument. After this limit has been generated, the data sequence will repeat. A great deal of system performance information can be deduced from the eye pattern display. To interpret the eye pattern, follow the given procedure.

The optimum sampling time corresponds to the maximum eye opening. ISI at this time partially closes the eye and thereby reduces the *noise margin*. If synchronization is derived from the zero crossings, as it usually is, zero- crossing distortion produces jitter and results in non optimum sampling times. The slope of the eye pattern in the vicinity of the zero crossings indicates the sensitivity of timing error. Finally, any nonlinear transmission distortion would reveal itself in an asymmetric or “squinted” eye.

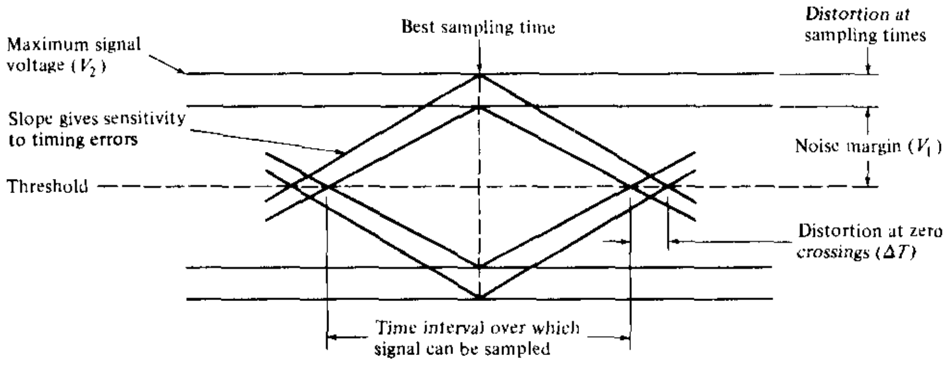
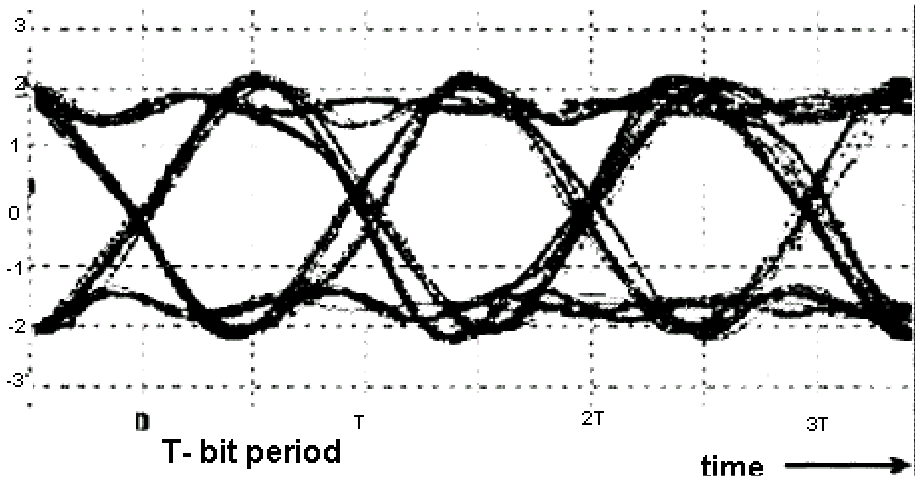
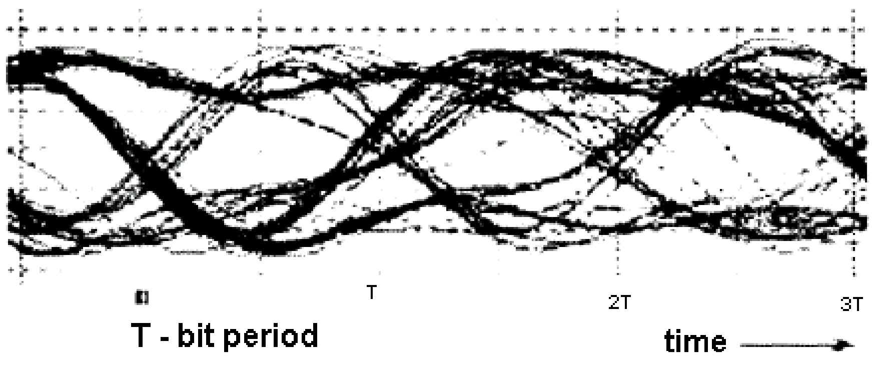


Fig.5.2 Simplified eye pattern diagram.



A 'good' eye pattern



A faster data rate

CALCULATIONS

NOISE MARGIN PERCENTAGE

The height of the eye opening at the specified sampling time shows the noise margin or immunity to noise.
Noise margin is the percentage ratio of the peak signal voltage V_1 for an alternating bit sequence (defined by the height of the eye opening) to the maximum signal voltage V_2 as measured from the threshold level.
That is:
Noise margin (percent) = $V_1/V_2*100\%$
Where,
V_1 = Noise margin.
V_2 = Maximum signal voltage.
TIMING JITTER:
Timing jitter (also referred to as edge jitter or phase distortion) arises from the noise at the receiver side. If the signal is sampled in the middle of the time interval (i.e. midway between the times when the signal crosses the threshold level) then the amount of distortion ΔT at the threshold level indicates the amount of jitter.
Timing jitter is thus given by:
Timing jitter (percent) = $\Delta T/T_b*100\%$
Where,
ΔT = Distortion at zero crossing.
T_b = One bit interval.

EQUIPMENTS

- DCL-BBT kit
- Connecting Chords
- Power supply
- 20MHz Dual Trace Oscilloscope

NOTE: Keep The Switch Faults In Off Position**.**

PROCEDURE

1. Refer to the block diagram (Fig.4.1) and carry out the following connections and switch settings.
2. Connect power supply in proper polarity to the kit **DCL-BBT** and switch it ON.
3. Keep the switch **SW1** as shown in the Fig.5.1. To generate PRBS signal.
4. Connect the data **OUT** post of PRBS generator to the **TX IN** post of the TX filter.
5. Set the rotary switch **SW2** of the TX channel to position no. **5**.
6. Connect the data from the **TX OUT** post to the **DATA IN** post of the Adder.
7. Connect the noise from the **NOISE OUT** post of the noise source to the **NOISE IN** post of the Adder.
8. Keep the noise level to minimum by rotating the pot **P1** in the anti clockwise direction.
9. Connect the **CLOCK OUT** post of PRBS generator to **EXT. TRIG** of oscilloscope.
10. Switch ON the **EXT. TRIG** button of the oscilloscope.

11. Connect the detected signal at the Adder **OUT** post to the vertical channel **Y** input of the oscilloscope.
12. Observe the eye pattern on the oscilloscope for different clock frequencies starting from 16 KHz to 1.024MHz.

Noise Margin And Timing Jitter Percentage:

13. To calculate the noise margin percentage and timing jitter percentage, set the clock frequency to 16 KHz and bandwidth position of the TX filter to position no.**1**.
14. Increase the noise level to maximum; perform the calculation as per the given formula.

    Noise margin percentage = $V_1/V_2*100\%$

    Timing jitter percentage = $\Delta T/T_b*100\%$
15. Increase the clock frequency to 32 KHz, and later to 128 KHz keeping the bandwidth position, to position no. **1** and perform the calculation as per the given formula.
16. Observe the differences in the percentage at different clock frequencies.
17. Keep the bandwidth position to 5 and the clock frequency of the PRBS generator to 16 KHz.
18. Now increase the clock frequency simultaneously and observe the opening of the eye.

OBSERVATION

Observe the eye pattern at the given clock frequencies, calculate the noise margin percentage and timing jitter percentage at different clock frequencies.

Clock frequency	Bandwidth position
1. 16 KHz	Position 1, 3, 5
2. 128 KHz	Position 2, 4, 5
3. 1.024 MHz	Positions 5

The width of the eye opening defines the time interval over which the received signal can be sampled without error from the inter-symbol interference. The best time to sample the received waveform is when the height opening is the largest.
The height of the eye opening is reduced as a result of amplitude distortion, it is given by the vertical distance between the top of the eye opening and the maximum signal level. The greater the eye closure becomes, the more difficult it is to detect the signal.
The rate at which the eye closes as the sampling time is varied (i.e. the slope of the eye pattern slides) determines the sensitivity of the system to timing errors. The possibility of the timing error increases as the slope becomes more horizontal.

CONCLUSION

It is observed that as the clock frequency increases the eye opening becomes smaller. With the increase of clock frequencies and the noise the noise margin percentage decreases which means that it becomes less immune to noise With the increase of noise in the data, the amount of distortion or jitter percentage increases.

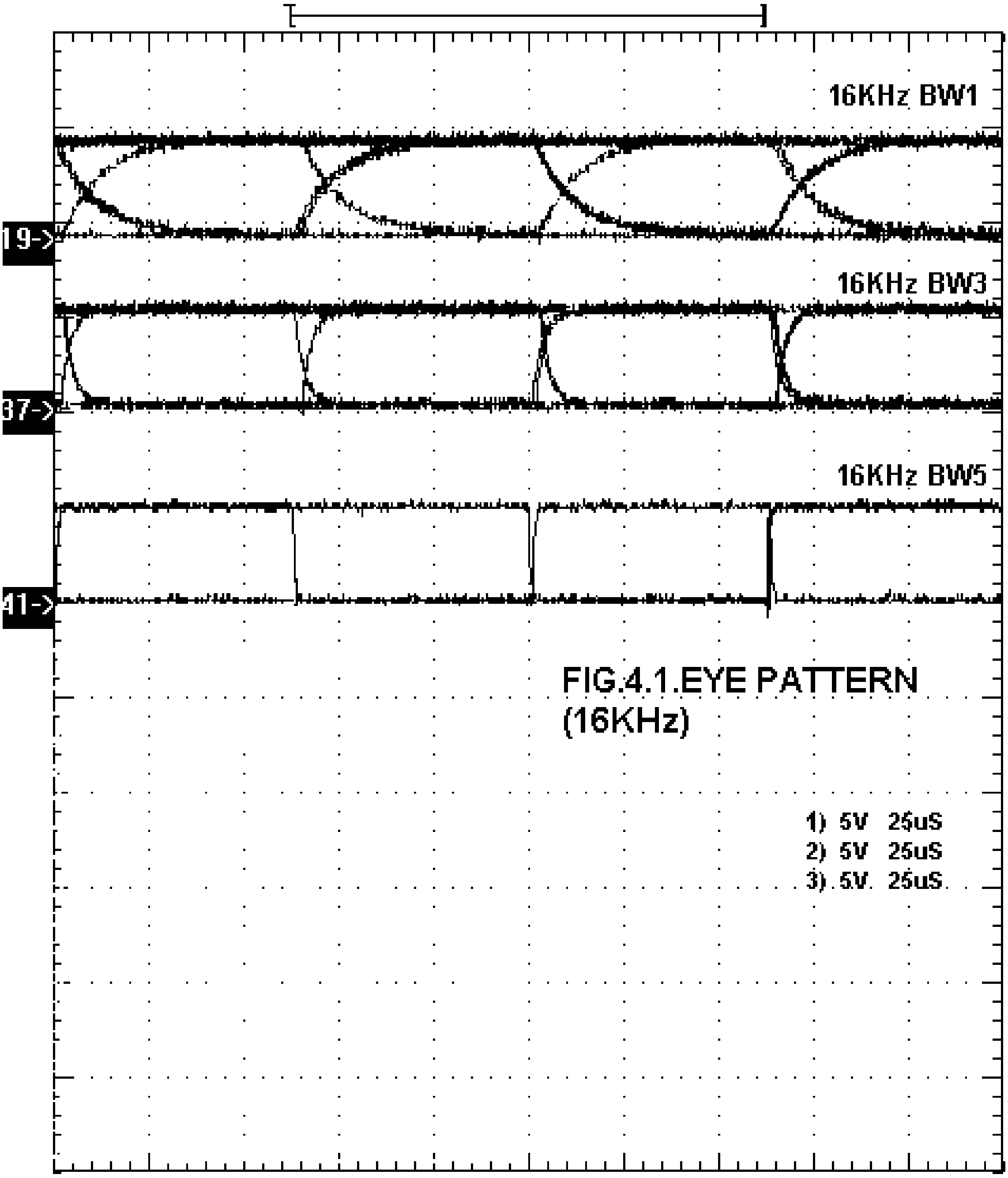


FIG.4.1.EYE PATTERN (16KHz)

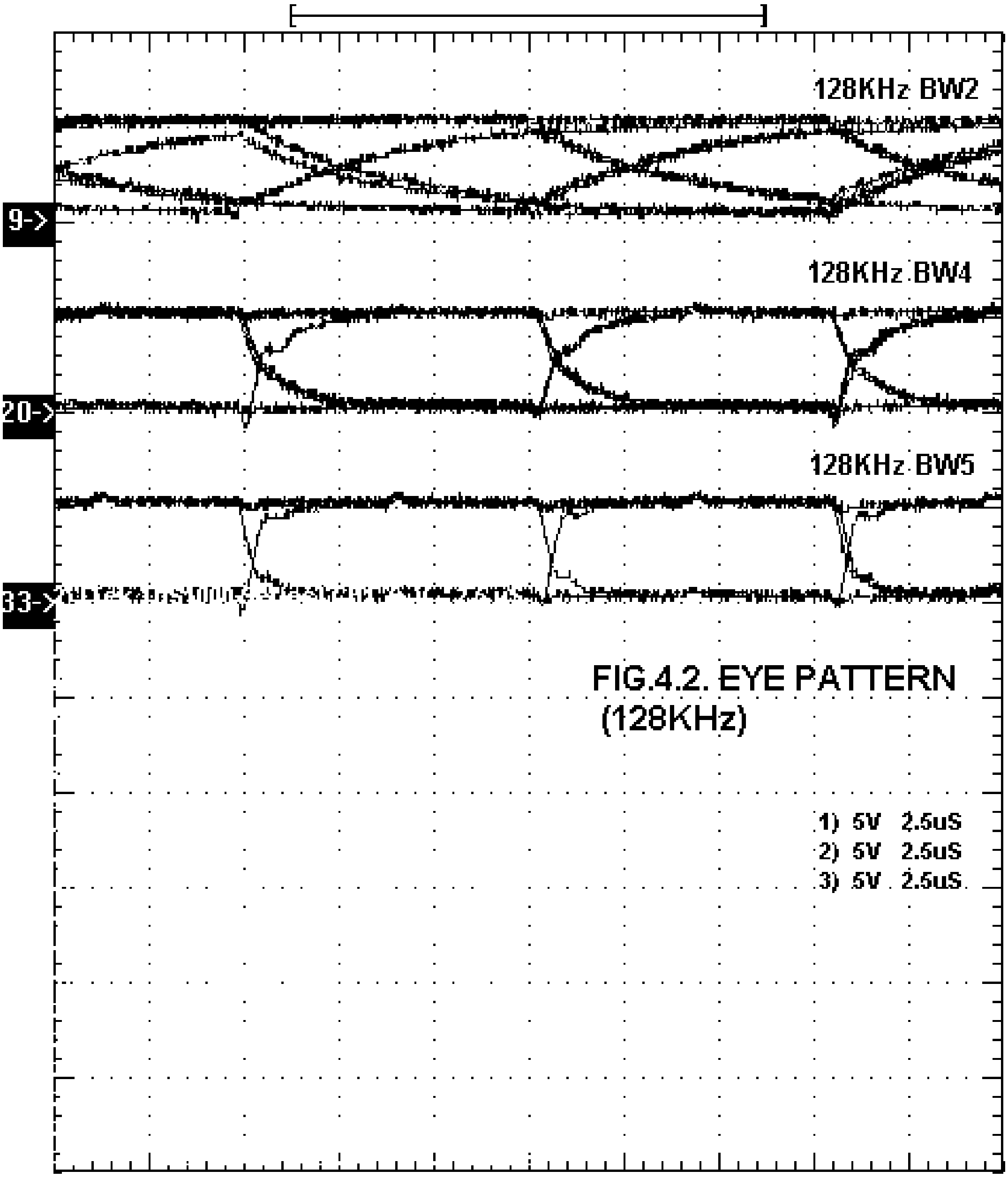


FIG.4.2. EYE PATTERN (128KHz)

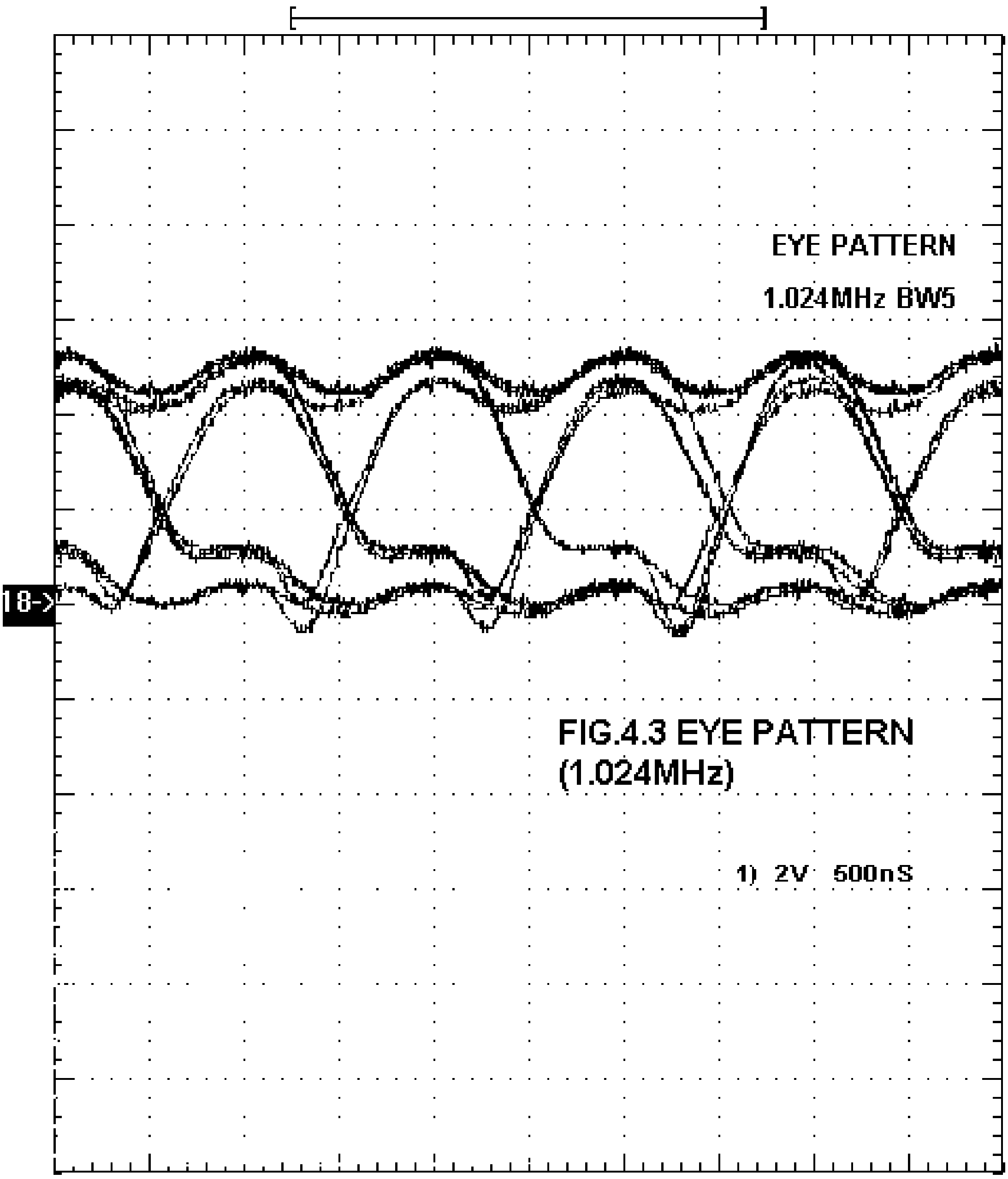
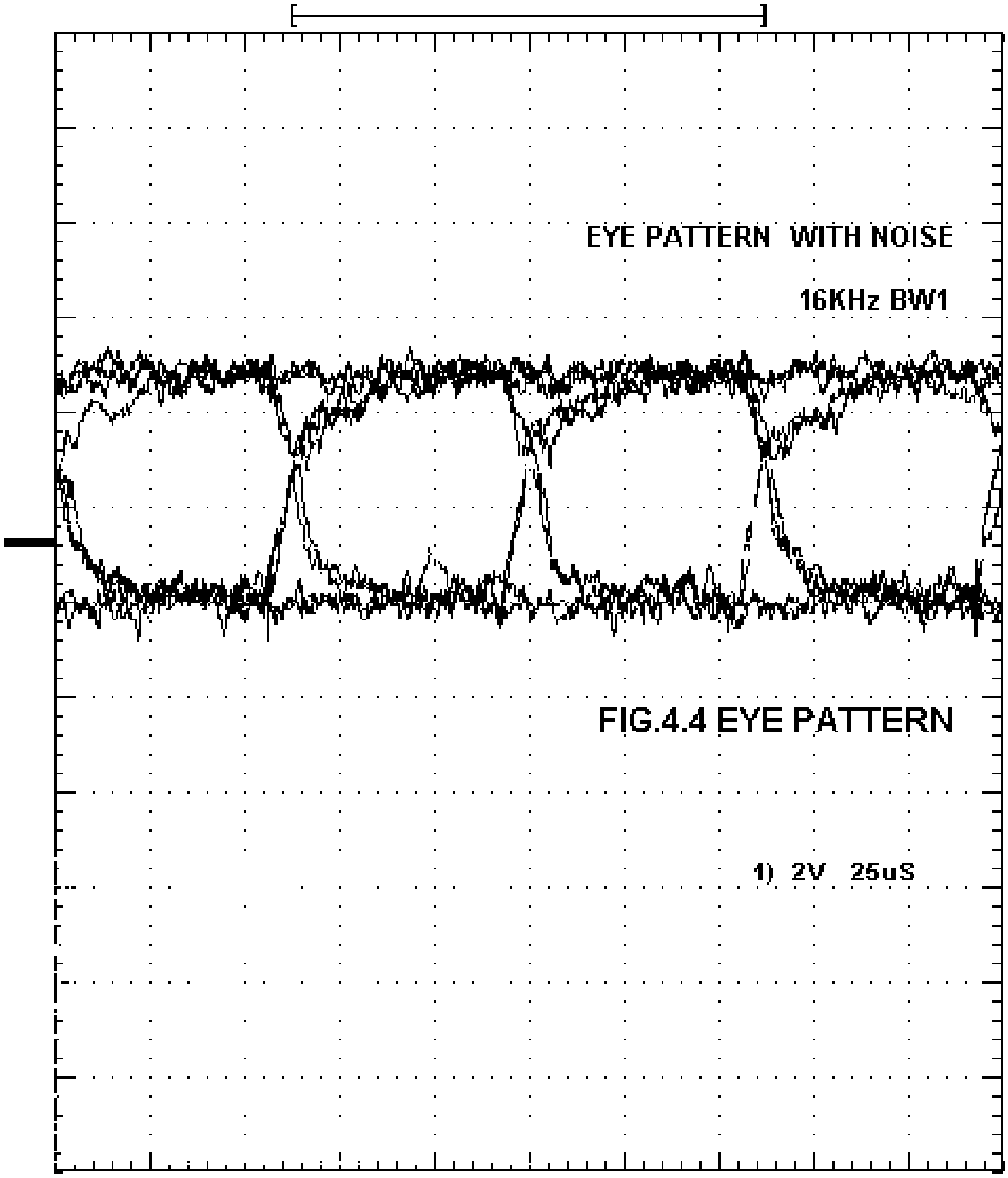


FIG.4.3 EYE PATTERN (1.024MHz)


EYE PATTERN WITH NOISE
16KHz BW1
FIG.4.4 EYE PATTERN
1) 2V 25uS

EXPERIMENT NO. 5

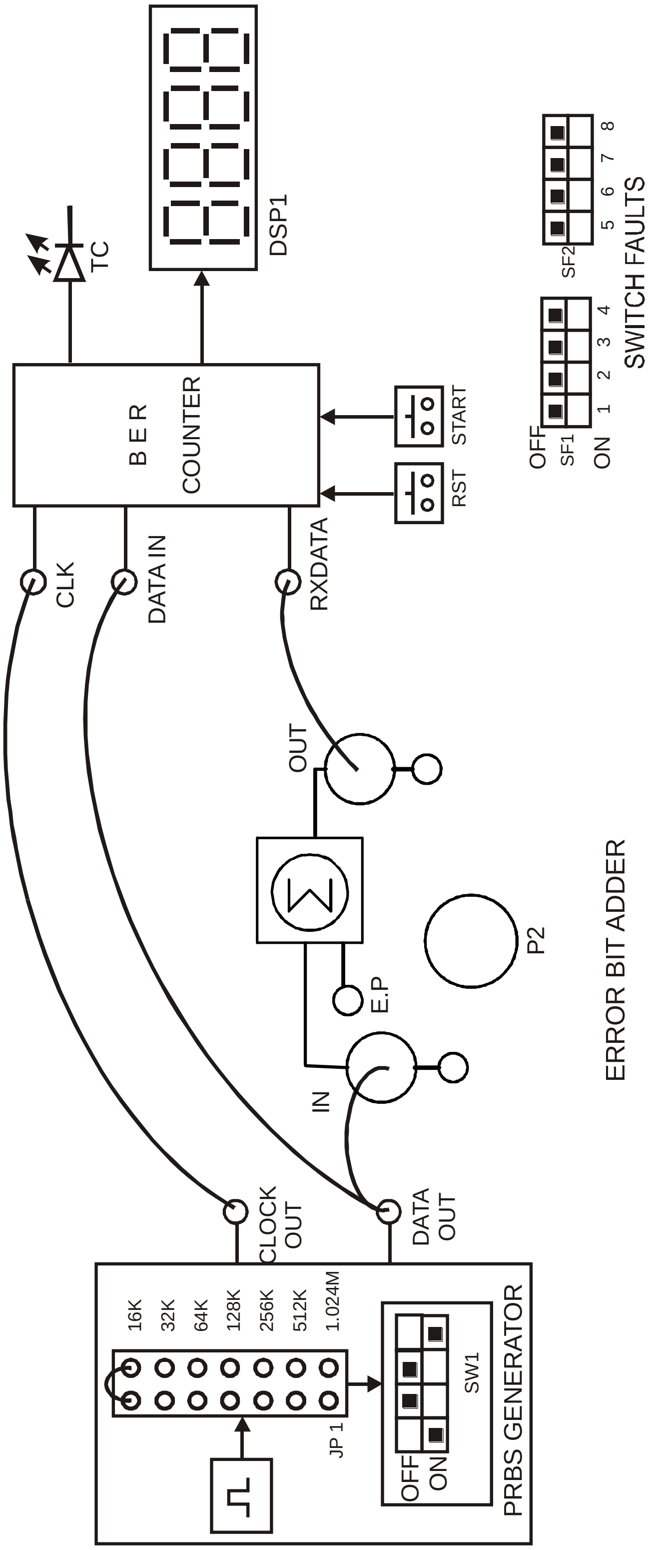


FIG. 5.1 BLOCK DIAGRAM FOR MEASUREMENT OF BIT ERROR RATE USING BINARY DATA

EXPERIMENT NO: 5

NAME

Measurement of Bit Error Rate Using Binary Data

OBJECTIVE

To measure bit error rate.

THEORY

In telecommunication transmission, the bit error rate (BER) is a ratio of bits that have errors relative to the total number of bits that have errors relative to the total number of bits received in a transmission. The BER is an indication of how often a packet or other data unit has to be retransmitted because of an error. Too high a BER may indicate that a slower data rate would actually improve overall transmission time for a given amount of transmitted data since the BER might be reduced, lowering the number of packets that had to be resent.

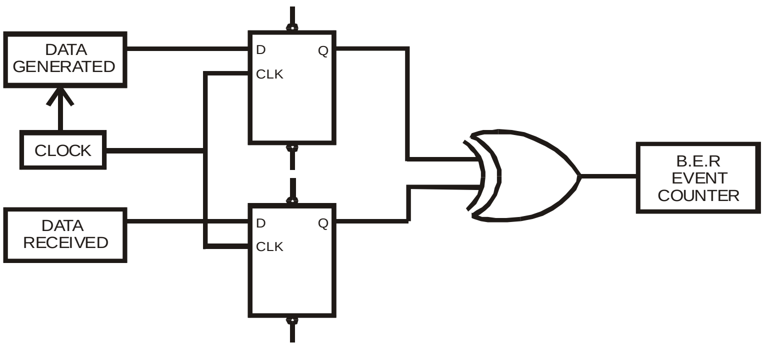
Measuring Bit Error Rate

A BERT (Bit Error Rate Tester) is a procedure or device that measures the BER for a given transmission. The BER, or quality of the digital link, is calculated from the number of bits received in error divided by the number of bits transmitted.

BER = (Bits in error) / (Total bits transmitted)

Using a bench test setup, this is easily measured by means of a comparator in which the transmitted bits are matched in an XOR gate with the received bits.

Fig shows the schematic of the device used for the following measurements.



BIT ERROR RATE TESTER (B.E.R.T.)

If the bits are alike at the XOR gate inputs when clocked in from the D flip-flop, the output is low. If they are different, the XOR output goes high, causing an event count.

A random character generator and white noise source should be used for these measurements.

The number of bit errors is dependent upon the amount of noise entering the system.

The white noise, or background noise, has an average or RMS value that is exceeded periodically by peaks that rise many times that level. This peak exists only for a short period of time. When the peak equals or exceeds the signal level, that is noise energy = bit energy, there is a 50/50 chance of error. The peak time periods can be calculated statistically from the error function.

EQUIPMENTS

- DCL-BBT kit
- Connecting Chords
- Power supply
- 20MHz Dual Trace Oscilloscope

NOTE: Keep The Switch Faults In Off Position.

PROCEDURE

1. Refer to the block diagram (Fig.5.1) and carry out the following connections and switch settings.
2. Connect power supply in proper polarity to the kit **DCL-BBT** and switch it on.
3. Keep the position of the switch **SW1** of the **PRBS GENERATOR** as shown in the Fig.5.1 for the generation of the PRBS pattern.
4. Select the **PRBS** clock frequency to **16 KHz** using jumper **JP1**.
5. Connect the PRBS **DATA OUT** post to the **IN** post of **ERROR BIT ADDER.**
6. Observe the noise pulses at **E.P.** by varying the pot **P2.**
7. Connect the **OUT** post of the Adder to the **RX DATA** post of the BER meter.
8. Keep the pot **P2** of the **ERROR BIT ADDER** at minimum position.
9. Connect the **CLK OUT** post to the **CLK** post of the BER meter.
10. Connect the **DATA OUT** post of **PRBS GENERATOR** to **DATA IN** post of the **BER METER**.
11. Press the **RST** switch to reset the counter to zero.
12. Press the **START** switch to start the counting.
13. Observe the error count on the display after 10 seconds upon pressing **START** switch.
14. Observe that when the pot is kept at minimum position the bit error count is very low. The error count will increase linearly as we increase the pot to the maximum (clockwise) position.

BER MEASUREMENT

The noisy data from the OUT post of the Adder is connected to the RX DATA IN of the BER meter As BER is the ratio of error bits (E_b) to total bits transmitted (T_b) in a period time *t* sec i.e.

$$BER = E_b/T_b$$

E.g. If PRBS data is transmitted at 32Kbits per sec for a period of 10 sec.

The total bits transmitted in 10 sec (T_b) = 320 bits.

The TTL out data with noise is fed to BER counter, which compares the two data input at each clock input.

The counter gives the 10 bit binary from error count (Eb) which is converted into decimal form and displayed.

E. g.

(0000001010 = 10 (decimal form))

BER RATIO then becomes:

BER = 10/320*$10E^3$

= 0.00003125

i.e. the channel Bit Error Rate ratio is 3.1*$10E^{-5}$ (3/100000) or in other words we can say that out of 100000 bits transmitted through the channel, the channel gives 3 bits in error.

EXPERIMENT NO. 6

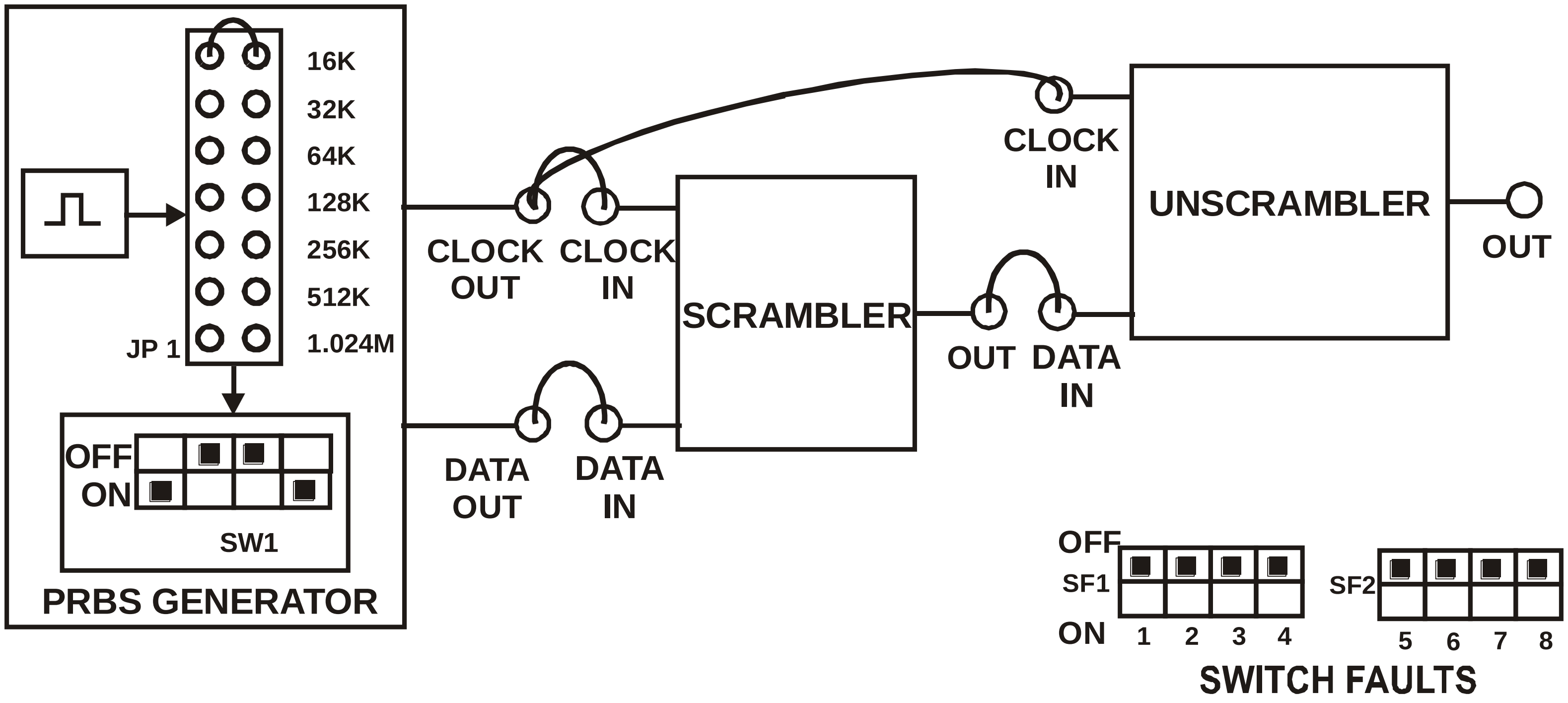


FIG. 6.1 BLOCK DIAGRAM FOR STUDY ON MESSAGE SCRAMBLERS AND UNSCRAMBLERS

EXPERIMENT NO: 6

NAME

Study on Message Scramblers and Unscramblers.

OBJECTIVE

To study the concept of scrambling and unscrambling of data

THEORY

An additional problem arises when the message includes long strings of 1's and 0's and the synchronism is lost during the working of synchronization techniques like in bit synchronization technique. Therefore message scramblers and unscrambles are introduced to help alleviate the problem.

SCRAMBLING

Scrambling is a coding operation applied to the message at the transmitter that "randomizes" the bit stream, eliminating long strings of like bits that might impair receiver synchronization. Scrambling also eliminates most periodic bit patterns that could produce undesirable discrete frequency components in the power spectrum. Needless to say, the scrambled sequence must be unscrambled at the receiver so as to preserve all bit sequence transparency. Simple but effective scramblers and un-scramblers are built from shift registers.

Message Scrambling And Unscrambling

The binary message sequence m_k at the input to the scrambler is mod-2 added to the register output m''_k to form the scrambled message m'k which is also fed back to the register input.

Thus,

$$m''_k = m'_{k-3} \oplus m'_{k-4}$$

$$m'_k = m_k \oplus m''_k \qquad \text{(a)}$$

The un-scrambler has essentially the reverse structure of the scrambler and reproduces the original message sequence, since

$$\begin{aligned} m'_k \oplus m''_k &= (m_k \oplus m''_k) \oplus m''_k \\ &= m_k \oplus (m''_k \oplus m''_k) \\ &= m_k \oplus (0) \\ &= m_k \qquad \text{(b)} \end{aligned}$$

Equation (a) and (b) hold for any shift register configuration as long as the scrambler and un-scrambler have identical registers.

EQUIPMENTS

- DCL-BBT kit
- Connecting Chords
- Power supply
- 20MHz Dual Trace Oscilloscope

NOTE: Keep The Switch Faults In Off Position.

PROCEDURE

1. Refer to the block diagram (Fig.6.1) and carry out the following connections and switch settings.
2. Connect power supply in proper polarity to the kit **DCL-BBT** and switch it on.
3. Keep the switch **SW1** as per the position shown in the block diagram fig.6.1 to generate the PRBS data pattern.
4. Connect the **CLK OUT** post of the PRBS generator to the **CLK IN** post of the Scrambler.
5. Connect the **DATA OUT** post to the **DATA IN** post of the Scrambler.
6. Connect the scrambled data from the **OUT** post of the Scrambler to the **DATA IN** post of the Un-scrambler.
7. Connect the **CLK OUT** post of the PRBS generator to the **CLK IN** post of the Un-scrambler.
8. Observe the output data for different clock rates and different data patterns.

OBSERVATION

- Scrambled data at the **OUT** post of the Scrambler on CH1 in dual channel with **DATA IN** on CH2.
- Unscrambled data at the **OUT** post of the Un-scrambler.

CONCLUSION

There is no loss or delay of data in the process of scrambling and unscrambling the binary data’s.